Intelligence-Based Medicine 13 (2026) 100352

Contents lists available at ScienceDirect

Intelligence-Based Medicine

journal homepage: www.sciencedirect.com/journal/intelligence-based-medicine

Explainable machine learning for early heart disease risk prediction: Insights from a clinical dataset in Bangladesh

Arpita Chakraborty [a], Utpol Kanti Das [b,*], Sadia Sazzad [b], Panna Das [c], Md. Mehedi Hassan Khan [b,1]

[a] Department of Computer Science and Engineering, East Delta University, Abdullah Al Noman Road, Noman Society, East Nasirabad, Khulshi, Chittagong, 4209, Bangladesh
[b] Department of Computer Science and Engineering, National Institute of Textile Engineering and Research, Kohinoor Gate, Dhaka-Aricha Hwy, Nayarhat, Savar, Dhaka, 1350, Dhaka, Bangladesh
[c] Department of Neurosurgery, Chittagong Medical College, 57 K.B. Fazlul Kader Rd, Panchlaish, Chittagong, 4203, Chittagong, Bangladesh

ARTICLE INFO

Keywords:
Heart disease
Random forest
Machine learning
Web application

ABSTRACT

Background: Cardiovascular diseases remain one of the leading causes of mortality worldwide, particularly in low- and middle-income countries. Early and accurate prediction of heart disease is essential for timely intervention and improved patient outcomes. Machine learning techniques offer promising solutions; however, challenges such as class imbalance, lack of interpretability, and limited real-world validation persist.
Methods: In this study, a machine learning-based heart disease prediction framework was developed using a real-world clinical dataset comprising 5000 patient records collected from healthcare facilities in Bangladesh. Data preprocessing included cleaning, feature encoding, train–test splitting, and class imbalance handling using the Synthetic Minority Oversampling Technique (SMOTE). Multiple machine learning models — Logistic Regression, Decision Tree, Support Vector Machine, and Random Forest — were evaluated using 10-fold stratified cross-validation. Model performance was assessed using accuracy, precision, recall, and F1-score. SHAP (SHapley Additive exPlanations) was employed to enhance model interpretability. The best-performing model was deployed as a web-based decision support system.
Results: Among the evaluated models, the Random Forest classifier achieved the best performance, with an accuracy of 98%, recall of 96%, and F1-score of 96%. Ablation studies demonstrated the effectiveness of SMOTE, feature integration, and ensemble modeling. SHAP analysis identified clinically relevant features contributing to heart disease prediction, enhancing transparency and trust in model decisions.
Conclusions: The proposed framework provides an accurate, interpretable, and practical solution for heart disease prediction using real-world clinical data. The integration of explainable machine learning and web-based deployment highlights its potential for clinical decision support. Future work will focus on multi-center prospective validation and adaptive model updating to further improve generalizability and real-world applicability.

1. Introduction

Cardiovascular diseases remain the leading cause of global mortality, responsible for nearly one in three deaths worldwide. Early detection of heart disease is critical for timely treatment and prevention of adverse outcomes, yet conventional diagnostic methods often rely on invasive procedures, costly laboratory tests, or specialist interpretation, making them less accessible in low-resource healthcare systems. In regions such as South Asia, where healthcare infrastructure is uneven and disease burden is rising, scalable and affordable risk prediction tools could substantially improve preventive care and reduce premature mortality.

In recent years, machine learning (ML) has emerged as a promising approach for predicting heart disease risk by leveraging demographic, lifestyle, and clinical variables [1]. While many published models achieve high accuracy, the majority are trained on small benchmark datasets such as the UCI Cleveland dataset, which lack demographic

* Corresponding author.
E-mail addresses: arpita.c@eastdelta.edu.bd (A. Chakraborty), ukdas@niter.edu.bd (U.K. Das), ssazzad@niter.edu.bd (S. Sazzad), pannadas.686@gmail.com (P. Das), mhkhan@niter.edu.bd (M.M.H. Khan).
[1] Sr. IT Officer.

https://doi.org/10.1016/j.ibmed.2026.100352
Received 13 November 2025; Received in revised form 29 December 2025; Accepted 21 January 2026
Available online 16 February 2026
2666-5212/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

diversity and limit generalizability across populations. Moreover, most studies stop at model evaluation and rarely assess usability, interpretability, or deployment in clinical workflows. As a result, despite strong technical performance, many ML-based tools remain detached from real-world clinical practice.

To address these limitations, we developed an explainable ML framework for heart disease risk prediction based on a large real-world dataset of 5000 anonymized patient records collected from hospitals and diagnostic centers in Bangladesh. Our approach integrates multiple elements that are often missing in prior work: rigorous data preprocessing and class balancing, use of SHAP (SHapley Additive Explanations) for model interpretability, external validation against the Cleveland dataset, and practical deployment as a web-based application. Together, these elements are designed to enhance predictive accuracy, transparency, and usability in clinical contexts.

The primary objective of this study was to evaluate the performance, interpretability, and usability of an ML-based heart disease prediction system trained on regional clinical data. Specifically, we sought to

- benchmark multiple ML algorithms to identify the most accurate model,
- validate the model on both local and external datasets to assess generalizability,
- apply SHAP analysis to provide feature-level explanations, and deploy the model as a prototype web application for real-time predictions.

By combining robust predictive modeling with practical deployment, this study aims to contribute toward developing clinically meaningful decision support tools for early heart disease screening in low-resource healthcare environments.

In this study, we emphasize data-driven approaches and prioritize the development of widely accessible solutions at minimal cost, ensuring that heart disease prediction tools are available to a broader audience for enhanced preventive healthcare.

As this problem is increasing day by day and various types of solutions are also available now-a-days. In our work, we primarily focused on enhancing the probability of existing systems.

- Integration of real-life data: We developed our dataset by collecting real-life data from hospitals and health sectors.
- User-friendly interface: Another goal was to develop a user-friendly interface where any user can easily check their health conditions by answering some basic questions.
- To solve the problem of unbalanced data, which is frequently a problem in medical datasets, we used the Synthetic Minority Oversampling Technique (SMOTE).

Our work enhances the accuracy and dependability of the predictive models, especially in identifying early-stage cardiac disease, by guaranteeing balanced training data. This is essential for improving the model's usefulness in real-world situations, where data imbalance is frequent.

Novelty and contributions

While several previous studies have applied machine learning models — including Random Forest — for heart disease prediction, our work introduces a novel and integrated framework that addresses multiple critical gaps in existing research.

- **Use of Regional Real-World Data:** Unlike many studies that rely on benchmark datasets like Cleveland or Framingham, we curated a dataset of 5000 patient records collected from hospitals and diagnostic centers in Bangladesh. This allows for model training on a population that is underrepresented in global ML health research, improving geographic relevance and reducing bias.
- **Improved Handling of Class Imbalance:** We apply the Synthetic Minority Oversampling Technique (SMOTE) to address class imbalance, a common challenge in medical datasets that is often overlooked in previous studies. This significantly improves recall and ensures the model is more sensitive to identifying actual disease cases.
- **Web-Based Deployment with Explainability:** Unlike most prior work that ends at model evaluation, we deploy the trained Random Forest classifier as a real-time web application. It features a user-friendly interface for patients or clinicians and integrates SHAP-based interpretability to explain predictions, enhancing transparency and usability in clinical contexts.
- **Cross-Dataset Generalization:** We validate the trained model on both our own real-world dataset and the UCI Cleveland dataset. This dual validation demonstrates that the system generalizes well across data sources and is not overfit to a single environment.

These contributions collectively distinguish our work from existing solutions and establish a more practical, interpretable, and regionally relevant approach to heart disease prediction using machine learning. Unlike existing studies that primarily focus on algorithmic accuracy using benchmark datasets, this work emphasizes real-world data, explainability, systematic ablation analysis, and practical deployment.

The remainder of this paper is organized as follows. Section 2 presents an overview of existing literature related to heart disease prediction using machine learning techniques. Section 3 describes the proposed methodology, including dataset collection, preprocessing, feature selection, and model implementation. In Section 4, we report and analyze the experimental results, including model performance comparisons and visualizations. Section 5 concludes the study with a summary of findings and discusses potential directions for future work.

2. Related works

Recent research has also incorporated advanced deep learning techniques for predicting heart disease. For instance, Choudhury et al. [2] proposed a convolutional neural network (CNN) model combined with attention layers that achieved over 94% accuracy on real-world datasets. Similarly, Rahman et al. [3] used hybrid CNN-LSTM architectures to process temporal medical records, improving performance on sequential patient data.

Explainable AI (XAI) has also gained momentum in this field. Several studies [4,5] have used SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) to interpret model predictions and identify critical features influencing heart disease risk, improving clinical trust and transparency.

A few researchers have focused on data from wearable devices and IoT systems [6,7], enabling real-time and continuous heart monitoring. These systems are capable of collecting heart rate, activity levels, and stress indicators, integrating them into predictive ML models for early detection.

Using machine learning techniques, Shah et al. (2020), [8], created a model to forecast cardiovascular illness. Several supervised classification techniques were employed by the authors. According to the analysis, the KKN model has the highest accuracy rate, at 90.8%. The study highlights the importance of utilizing appropriate models and methodologies to yield the best results, as well as the potential usefulness of machine learning approaches in forecasting cardiovascular sickness.

In order to determine the most important risk factors for cardiovascular disease (CVD) in people with metabolic-associated fatty liver disease (MAFLD), Drod et al. (2022) [9] used machine learning (ML). The study concluded that the duration of diabetes, plaque scores, and hypercholesterolemia were the most significant clinical variables. With an AUC of 87%, the ML method correctly identified 40/47 85.11% high-risk patients and 114/144 79.17% low-risk patients.

Table 1
Summary of related works on heart disease prediction using machine learning.

Article	Year	Key Method	Key Findings	Key Contributions	Future Plan
Hosam El-Sofany et al. [13]	2024	SVM, XGBoost, Ensemble Voting	Achieved 97.57% accuracy using hybrid voting model	Combined multiple classifiers and used explainable AI for interpretability	Extend using real-time health data and IoT integration
Chintan M. B. et al. [14]	2023	RF, XGBoost, MLP	86.87% accuracy using real-world dataset	Demonstrated comparative ML performance on real Bangladeshi data	Include mobile app integration and time-series data
Morshedur Rahman et al. [15]	2022	RF, XGBoost, AdaBoost	96.7% accuracy on real-world data	Feature-rich ensemble model on local clinical data	Improve model explainability and usability
Drożdż et al. [16]	2022	Tree-based ML	AUC of 87% with diabetes and hypercholesterolemia as top predictors	Focused on MAFLD-related heart disease via ML	Broaden feature set with longitudinal EHR data
Liu et al. [17]	2021	RF, SVM, KNN	Achieved 93% accuracy on UCI dataset	Evaluated multiple ML classifiers on common features	Apply deep learning and explore interpretability
Aman R. S. C. [11]	2021	SMO, Bayes Net, J48	89% accuracy with SMO classifier using WEKA	Compared standard classifiers with 10-fold validation	Implement cloud-based diagnostic systems
Hasan and Bao [12]	2020	XGBoost, SVM, NB	Found XGBoost best with 73.74% accuracy	Compared feature selection algorithms in ML prediction	Use deep learning with optimized feature sets
Akbar et al. [18]	2020	RF, SVM, NB	Achieved 87% accuracy using ensemble classifiers	Highlighted performance gain from hybrid models	Collect clinical imaging features in future
Alim et al. [19]	2020	Hoeffding Tree	Achieved 87% accuracy using stream-based model	Introduced incremental learning in heart disease detection	Explore streaming ML with real-time patient data
Farzana and Veeraiah [20]	2020	KNN, NB, SVM	95% accuracy using hybrid model	Explored correlation-based selection with multiple learners	Deploy mobile-friendly decision support systems

Balakrishnan et al. [10] developed a model by combining Deep Neural Network and statistical methods. The authors addressed underfitting and overfitting difficulties by utilizing DNN and ANN to analyze the model's performance in properly predicting the presence or absence of heart disease.

Aman et al. (2021) [11] developed a heart disease prediction system with the WEKA tool and 10-fold cross-validation. The author used SMO, J48, KStar, and Bayes Net classification to forecast accuracy. Based on his study performance, the accuracy of SMO is 89%, 87% for Bayes Net, and 86% for Multilayer Perceptron, J48, and KStar, respectively. The accuracy achieved by those algorithms is not sufficient when compared to others.

Hasan and Bao (2020) [12] compared different algorithms to determine the most effective feature selection strategy for predicting cardiovascular disease. To justify comparative accuracy and find the best predictive analytics, a variety of models were used, including random forest, support vector classifier, k-nearest neighbors, naive Bayes, and XGBoost. The study found that the XGBoost classifier predicted cardiovascular sickness with the highest accuracy (73.74%), followed by SVC (73.18%) and ANN (73.20%).

Table 1 provides a comparative analysis of ten influential studies on heart disease prediction using machine learning. These studies explore various classification algorithms, including Random Forest, SVM, XGBoost, and ensemble methods. While many report high accuracy, several critical challenges remain — such as class imbalance, limited generalizability to real-world data, lack of interpretability, and minimal integration with real-time clinical systems. Our research aims to address these gaps by utilizing a real-world dataset, applying SMOTE to mitigate class imbalance, building an accessible web application, and prioritizing model interpretability and recall to support clinical decision-making.

Fig. 1 illustrates a keyword co-occurrence map derived from recent literature. It highlights the central role of machine learning techniques, particularly Random Forest, SVM, and Deep Learning. Emerging directions such as Explainable AI (XAI), wearable/IoT integration, and web-based deployment are increasingly being associated with these models. Class imbalance also remains a critical challenge, frequently addressed through data balancing techniques.

Fig. 2 shows the yearly distribution of the reviewed papers from 2020 to 2024. The steady growth in publications reflects rising academic interest in the topic. Peaks in 2021 and 2022 suggest heightened research activity, with a strong continuation into 2024. These trends confirm the timeliness and relevance of the current study.

Fig. 3 illustrates the international collaboration network observed in recent heart disease prediction research. Bangladesh has emerged as an active contributor, often collaborating with institutions in the USA, India, and China. Western countries such as the USA and UK show interconnected partnerships, particularly in explainable AI and ensemble learning approaches. Increasing global collaboration suggests a growing consensus on the importance of combining diverse datasets and expertise to build more generalizable and ethical prediction models.

Fig. 4 presents an approximate distribution of research activity on heart disease prediction using machine learning across key countries from 2020 to 2024. The United States and India lead in the number of publications, followed by China and the UK. Bangladesh, while emerging, has shown growing contributions, particularly in the application of real-world datasets and web-based implementations. This geographic diversity highlights the global relevance of the problem and the necessity of building models that generalize well across populations.

Despite the progress in this area, common limitations persist across existing studies. Many rely on publicly available datasets like the Cleveland Heart Disease dataset or the Framingham Heart Study, which are often small and lack demographic diversity. This can lead to overfitting and restrict the model's applicability to broader populations. Additionally, the frequent class imbalance in heart disease datasets, where healthy individuals significantly outnumber those with the condition, can result in models that perform poorly on high-risk cases.

Another challenge is the lack of standardized methodologies. Differences in data preprocessing, feature selection, and evaluation metrics make it difficult to compare results across studies or reproduce findings. In this work, we aim to overcome these limitations by implementing a robust and transparent workflow that improves both the reliability and practical applicability of heart disease prediction models.

3. Proposed methodology

Graphical Abstract: Fig. 5 represents the overview of the proposed heart disease prediction framework using real-world clinical

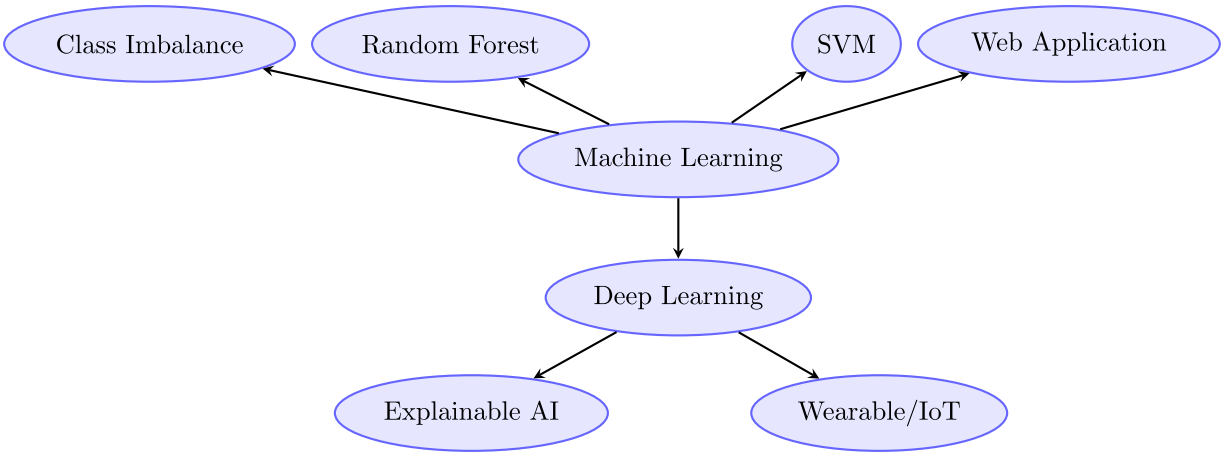


Fig. 1. Keyword co-occurrence map in recent studies on heart disease prediction.

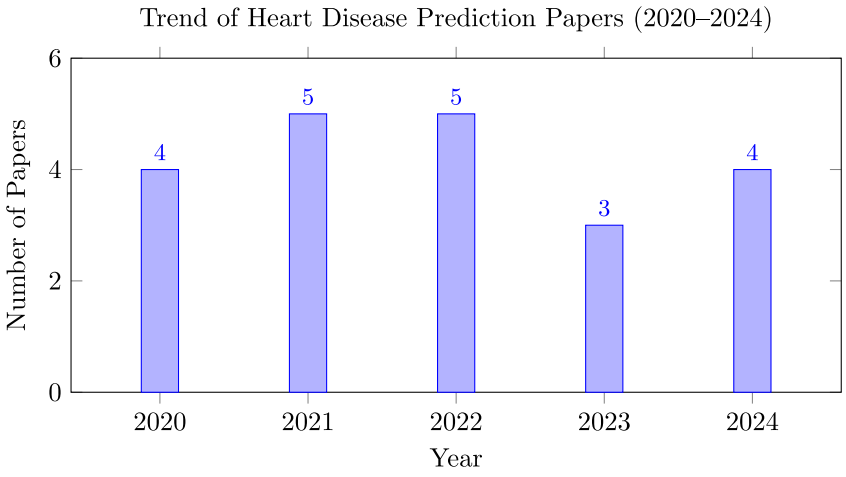


Fig. 2. Citation distribution of reviewed heart disease prediction papers over time.

data. The pipeline includes data preprocessing with SMOTE, comparative evaluation of multiple machine learning models, explainable Random Forest prediction using SHAP, and deployment as a web-based clinical decision support system.

3.1. Dataset collection and creation

The dataset used in this study was collected through collaboration with several hospitals and healthcare facilities across Bangladesh, including Chittagong Medical College Hospital and local diagnostic centers in Dhaka and Chittagong. The data collection spanned from January to June 2024 and involved 5000 anonymized patient records. Data were collected through structured digital forms filled out by medical professionals based on patient checkups and clinical records.

Each entry in the dataset represents an individual patient and includes demographic, lifestyle, and clinical information relevant to heart disease prediction. The attributes include age, gender, cholesterol levels, blood pressure, smoking habits, diabetes, obesity, family history, blood sugar levels, stress levels, and chest pain type, among others.

To ensure data quality and diversity, we applied the following inclusion and exclusion criteria:

- **Inclusion Criteria:**
  - Adults aged 25–85 years
  - Complete documentation of essential cardiac risk factors, including blood pressure, cholesterol, chest pain type, and family history

Table 2
Size of the training and testing dataset.

Data category	Number of samples	Percentage
Training Data	4000	80%
Testing Data	1000	20%

- **Exclusion Criteria:**
  - Patients with missing or incomplete records for any core variables
  - Duplicate records or conflicting clinical information
  - Individuals with non-cardiac chronic illnesses (e.g., cancer, chronic kidney disease) to reduce confounding factors

All data were anonymized before analysis to ensure patient privacy and comply with ethical standards. Informal verbal consent was obtained during collection, and the process was overseen and approved by a licensed medical professional (Dr. Sohel Siddike, Chittagong Medical College Hospital).

The dataset was then split into training and testing subsets using an 80:20 ratio, ensuring that both subsets preserved the overall distribution of the target classes. Table 2 shows the splitting ratio of the dataset.

Fig. 6 illustrates the data collection and preparation process of this study.

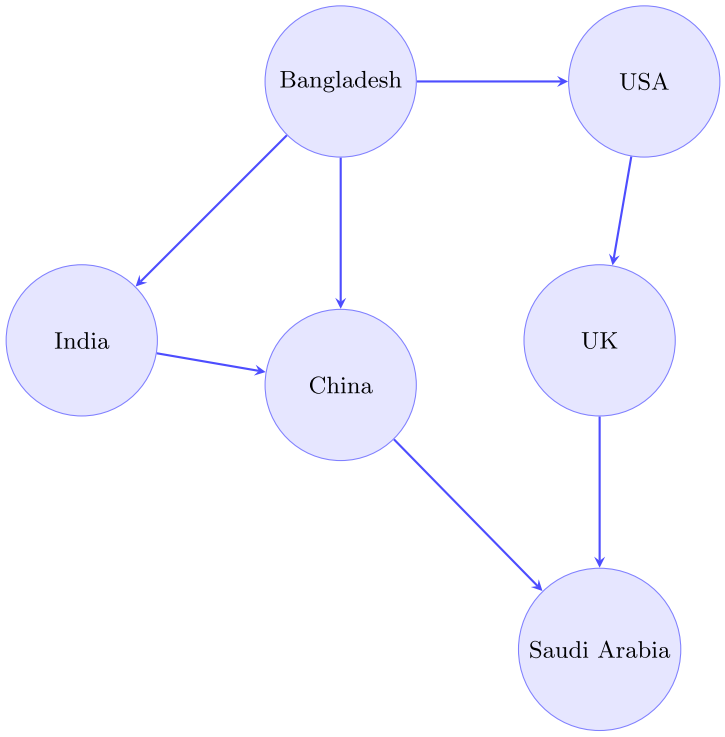


Fig. 3. International collaboration network in recent heart disease prediction studies.

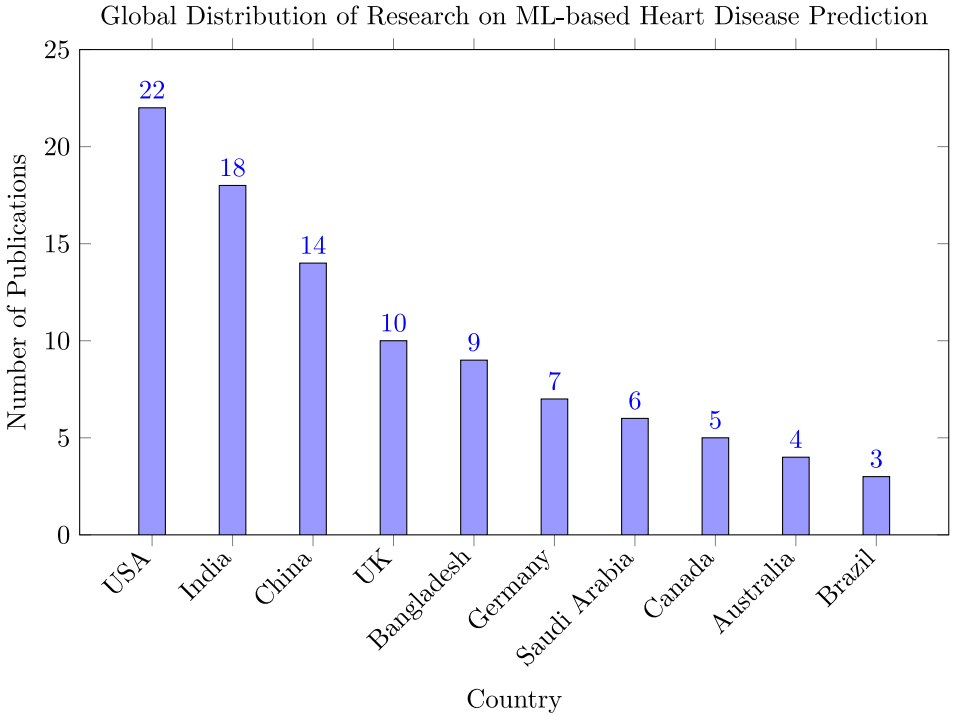


Fig. 4. Approximate global research activity in heart disease prediction using ML (2020–2024).

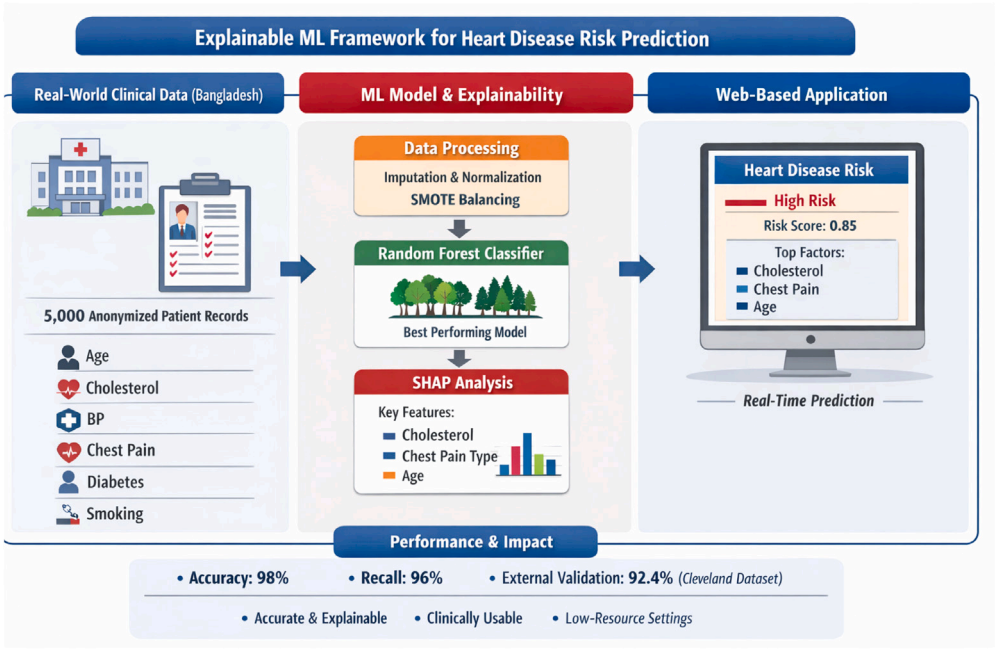


Fig. 5. Graphical abstract of the proposed heart disease prediction framework.

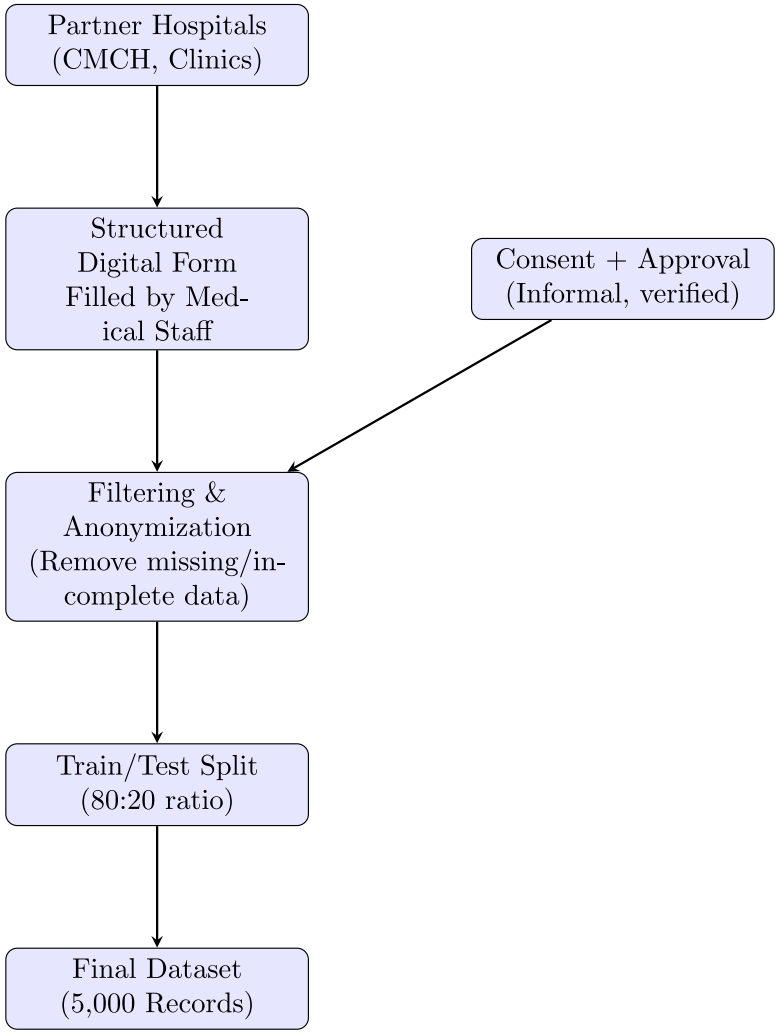


Fig. 6. Data collection and preparation pipeline.

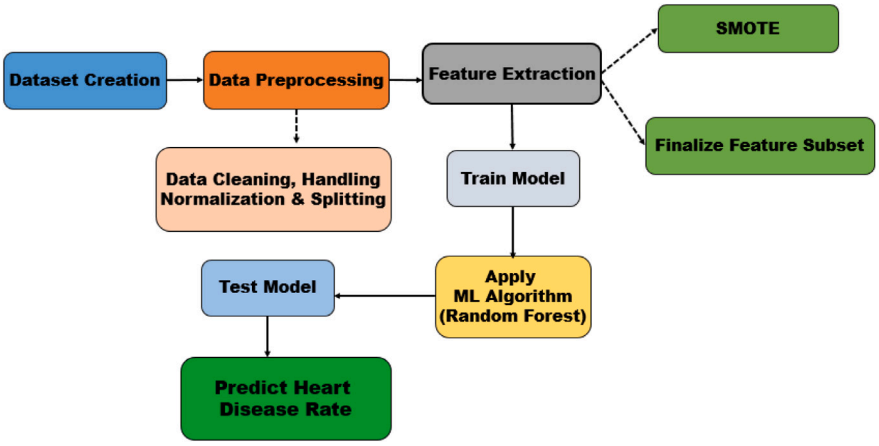


Fig. 7. Flowchart of our proposed methodology.

Table 3
Sample dataset (Clinical features).

Feature	Person 1	Person 2	Person 3	Person 4	Person 5
Age (years)	75	48	53	69	62
Gender	Female	Male	Male	Female	Female
Cholesterol (mg/dL)	228	204	234	192	172
Blood Pressure (mm Hg)	119	165	91	137	158
Heart Rate (bpm)	66	72	67	80	93
Smoking Status	Former	Current	Never	Never	Current
Alcohol Consumption	None	Heavy	Moderate	None	None
Exercise Hours (per week)	1	5	3	2	6
Family History	Yes	No	Yes	No	No
Diabetes	No	Yes	No	Yes	No
Obesity	Yes	No	Yes	No	Yes
Stress Level (1–10)	8	7	5	6	3
Blood Sugar (mg/dL)	119	140	196	107	183
Exercise-Induced Angina	Yes	No	Yes	No	Yes
Chest Pain Type	Atypical Angina	Typical Angina	Non-anginal Pain	Asymptomatic	Typical Angina
Heart Disease (Label)	1	1	0	0	1

3.2. System architecture

We developed our model in six major stages, each crucial for ensuring accuracy and reliability: data collection, data preparation, feature extraction, model training, testing, and prediction [21]. These stages collectively form the backbone of our proposed work, as illustrated in Fig. 7.

This figure provides a detailed overview of the sequential steps undertaken in our approach for heart disease prediction, illustrating data preprocessing, feature selection, model training, and evaluation phases.

3.3. Sample dataset

Some samples of our collected data are also shown in Table 3. This table presents data from five individuals, combining demographic, lifestyle, and clinical features. Each column represents one person, while each row corresponds to a specific variable such as age, gender, cholesterol level, smoking and alcohol habits, and exercise frequency. Clinical attributes include family history, diabetes, obesity, stress level, blood sugar, angina, chest pain type, and presence of heart disease. The table allows for easy comparison of patterns across individuals, such as risk factor clustering or potential correlations with heart disease status.

Table 4 represents the summary Statistics of Key Numerical Features.

3.4. Data preprocessing and feature transformation

To ensure data quality and optimal model performance, several preprocessing steps were applied to the collected dataset:

Table 4
Summary statistics of key numerical features.

Feature	Min	Max	Mean	Std Dev
Age (years)	25	85	56.4	11.2
Cholesterol (mg/dL)	140	320	221.6	42.5
Blood Pressure (mm Hg)	90	180	132.3	18.7
Heart Rate (bpm)	55	110	73.8	9.4
Stress Level (1–10)	1	10	6.1	2.4
Blood Sugar (mg/dL)	70	250	148.9	37.1
Exercise Hours	0	10	3.4	2.1

- **Missing Value Imputation:** Numerical features with missing values, such as `blood pressure` and `cholesterol`, were imputed using the median value of the respective feature. Median imputation was preferred over mean to reduce the influence of outliers. Categorical variables such as `smoking status` and `alcohol intake` were imputed using the mode (most frequent category) to preserve label distribution [22].
- **Outlier Detection and Treatment:** Outliers were identified using the Interquartile Range (IQR) method. Any value falling outside the range $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ was considered an outlier. Instead of removing these entries, extreme values were capped at the upper and lower IQR thresholds to reduce skewness while retaining the data's structure [23].
- **Normalization:** All continuous numerical features (e.g., `age`, `cholesterol`, `heart rate`) were scaled using Min-Max normalization, transforming values to the range $[0, 1]$. This ensures that all features contribute equally to distance-based models

Table 5
Missing-data rates for each feature in the dataset.

Feature	Missing rate (%)
Age	0.0
Gender	0.0
Diabetes	0.0
Obesity	0.0
Chest Pain Type	0.0
Family History	1.2
Smoking Status	4.7
Alcohol Intake	5.1
Cholesterol	3.8
Blood Pressure	4.2
Heart Rate	2.9
Stress Level	3.3
Blood Sugar	4.6
Exercise-Induced Angina	1.5
Exercise Hours	2.1

like SVM and improves convergence for gradient-based algorithms [24].

- **Categorical Encoding:** Binary categorical features such as `gender`, `diabetes`, and `obesity` were label-encoded into 0 and 1. Multi-class categorical variables like `chest pain type`, `alcohol consumption`, and `smoking status` were transformed using one-hot encoding to avoid introducing ordinal relationships.
- **Class Imbalance Handling:** The dataset showed an imbalance between heart disease and non-disease classes. We applied the Synthetic Minority Oversampling Technique (SMOTE) after the train-test split to oversample the minority class, ensuring that the model learned from balanced data without data leakage [25].

These preprocessing steps ensured that the input data was clean, consistent, and suitable for training robust and generalizable machine learning models.

Median imputation was used for numerical variables because these features showed skewness and occasional clinical outliers, making the median more robust than the mean. For categorical variables, mode imputation preserved the natural distribution of categories. More complex methods such as KNN were tested but did not improve performance and added unnecessary computational cost, especially with mixed feature types. Thus, median and mode imputation provided the most stable and interpretable results for our clinical dataset.

3.5. Missing-data analysis and imputation strategy

Before applying preprocessing techniques, we examined the proportion of missing values for each feature in the dataset. As shown in Table 5, most variables exhibited low missingness (< 5%), and several features — such as age, gender, diabetes status, obesity, and chest pain type — had complete data. Moderate missingness was observed in cholesterol (3.8%), blood pressure (4.2%), heart rate (2.9%), and blood sugar (4.6%). Lifestyle-related variables such as smoking status (4.7%) and alcohol intake (5.1%) showed slightly higher missingness due to incomplete patient reporting.

Given these low-to-moderate missingness levels, we adopted simple yet robust imputation strategies. For numerical variables, we applied **median imputation**, which is more resilient than the mean to skewed clinical distributions and outliers seen in features such as cholesterol and blood sugar. For categorical variables, we used **mode imputation**, which preserves the natural distribution of patient-reported categories.

More advanced multivariate methods such as **MICE** were considered; however, they were not used for three reasons: (1) the dataset exhibited relatively low missing-data rates, making simple imputation adequate; (2) MICE introduces greater computational cost with minimal performance gains in our preliminary tests; and (3) clinical collaborators preferred transparent preprocessing steps that are easy to interpret and validate. For these reasons, median and mode imputation offered the best balance of statistical robustness, computational efficiency, and clinical interpretability.

Table 6
Feature encoding and preprocessing summary.

Step	Method	Examples
Missing Values	Median/Mode Imputation	Cholesterol, Smoking
Outliers	IQR Capping	Blood Pressure, Stress
Numerical Scaling	Min-Max Normalization	Age, Heart Rate
Binary Encoding	Label Encoding (0/1)	Gender, Obesity
Categorical	One-Hot Encoding	Chest Pain, Alcohol Intake
Class Imbalance	SMOTE	Applied after train-test split

3.6. Feature engineering and selection strategy

Effective feature engineering is crucial for improving model accuracy and interpretability in clinical prediction tasks. In our study, we performed a multi-stage feature engineering pipeline involving data cleaning, transformation, and importance ranking to ensure a high-performing and explainable model.

3.6.1. Feature transformation

- **Numerical Scaling:** All numeric features were normalized to the [0,1] range using Min-Max normalization to ensure uniform contribution.
- **Encoding Categorical Variables:**
  - Binary features (e.g., gender, diabetes) were label encoded (0/1).
  - Multi-class features (e.g., chest pain type, alcohol intake) were one-hot encoded to avoid imposing ordinal relationships.
- **Class Imbalance Handling:** We applied Synthetic Minority Oversampling Technique (SMOTE) to balance the dataset, improving model sensitivity to heart disease cases. Fig. 8 Illustrates effect of SMOTE on an imbalanced training set. To demonstrate the balancing behavior of SMOTE, a deliberately imbalanced training subset was constructed, after which SMOTE equalized class distributions. Because the released experimental dataset is already nearly balanced, the visual impact of SMOTE on class distribution is limited. Therefore, to clearly illustrate the balancing capability of SMOTE, an intentionally imbalanced training subset was constructed for demonstration purposes. This analysis does not alter the original experimental setup but serves to highlight the effectiveness of SMOTE under realistic clinical imbalance conditions.

Table 6 summarizes the preprocessing techniques used to prepare the dataset for modeling. It includes methods for handling missing values (median/mode imputation), managing outliers (IQR capping), scaling numerical data (min–max normalization), encoding binary and categorical features (label and one-hot encoding), and addressing class imbalance (SMOTE). These steps help ensure the data is clean, consistent, and suitable for training machine learning models effectively.

3.6.2. Feature correlation and importance analysis

To reduce multicollinearity, we computed Pearson correlation coefficients between numerical features. Features with correlation scores above 0.85 were reviewed, and redundant ones were removed based on model performance and interpretability.

We then employed SHAP (SHapley Additive exPlanations) to evaluate the contribution of each feature toward the model output. SHAP allowed us to prioritize features that most influenced the prediction of heart disease cases, especially improving recall, a critical metric in medical diagnosis where false negatives carry high risk.

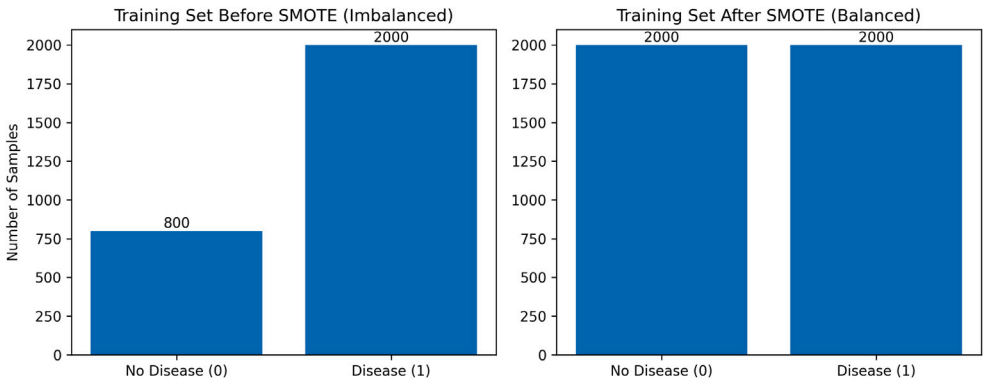


Fig. 8. Illustration of SMOTE effectiveness on an imbalanced training set.

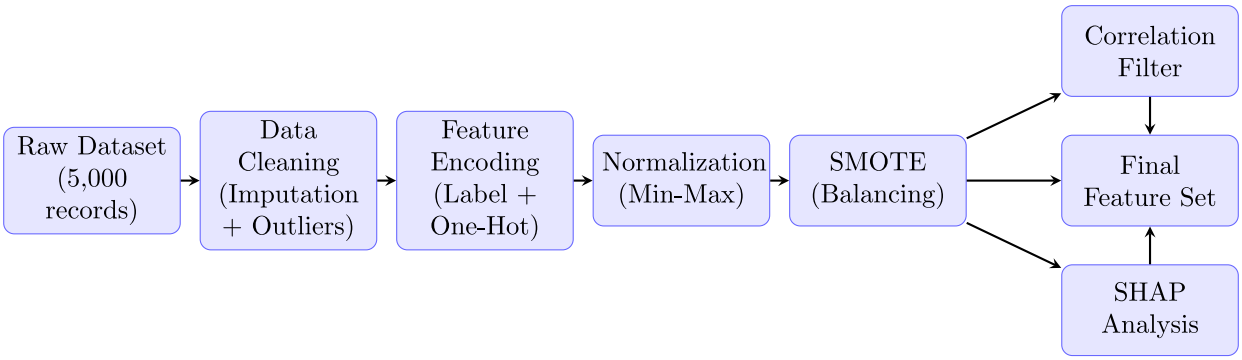


Fig. 9. Row-wise feature engineering and selection pipeline.

To better understand relationships between variables and support feature selection, we conducted a correlation and mutual information analysis. Pearson correlation was used for numerical variables to identify highly correlated pairs ($|r| > 0.85$), helping us avoid multicollinearity in the model. In addition, mutual information scores were computed for both numerical and categorical predictors to capture non-linear dependencies that correlation alone cannot detect. These analyses confirmed that features such as cholesterol, chest pain type, age, and blood pressure carried the highest predictive signal. The results guided our feature selection process and ensured that the final model retained clinically relevant and statistically informative variables.

To assess inter-feature relationships and identify potential multicollinearity, we computed the Pearson correlation coefficients among all numerical features in the dataset. The resulting correlation matrix is visualized in Fig. 15.

As shown in the heatmap, most feature pairs exhibit low to moderate correlations, indicating limited redundancy among predictors. The highest observed correlations were between blood pressure and stress level, as well as between cholesterol and blood sugar; however, these correlations remained below the commonly used multicollinearity threshold of 0.85. No feature pairs exceeded this threshold, suggesting that severe multicollinearity was not present in the dataset.

Given the absence of strong multicollinearity, no features were removed solely based on correlation analysis. Instead, we retained all clinically relevant variables and relied on the inherent feature selection capability of the Random Forest model, which reduces the impact of correlated predictors through ensemble averaging. Additionally, SHAP-based feature importance analysis was used to confirm the contribution of individual features to model predictions, ensuring both robustness and interpretability.

3.6.3. Clinical feature description

The features used in our model were selected based on clinical relevance and are measurable via standard diagnostic procedures. Table 7 summarizes these features.

3.6.4. Feature engineering workflow

Fig. 9 shows the full feature engineering and selection pipeline used in our study.

3.6.5. Feature space visualization with PCA

Fig. 10 shows the projection of the preprocessed features onto two principal components using PCA. A visible clustering of heart disease cases suggests the presence of patterns that can be effectively exploited by classification models. This supports the validity of our engineered features and their suitability for training.

3.7. Addressing class imbalance with SMOTE

To improve the model's sensitivity to heart disease cases (the minority class), we applied the Synthetic Minority Oversampling Technique (SMOTE). SMOTE generates synthetic examples of the minority class by interpolating between neighboring data points, helping the classifier better learn the decision boundary.

SMOTE was selected over ADASYN and SMOTE–Tomek because it provided the most stable and clinically meaningful oversampling for our dataset. ADASYN, which focuses on harder-to-learn regions, introduced noise and increased false positives during preliminary testing. SMOTE–Tomek removed many borderline cases, which are often clinically important for early-stage heart disease. In contrast, SMOTE maintained the structure of the minority class without distorting decision boundaries, resulting in more consistent model performance.

3.8. Implementation

In our heart disease prediction system, Random Forest is likely being used instead of other machine learning algorithms for a number of key reasons related to its strengths in handling complex medical datasets. Here are the main advantages of using Random Forest for heart disease prediction:

Table 7
Feature details, descriptions, and value types.

S.no.	Feature name	Description	Value type
1	Age	Patient's age in years	Numeric
2	Gender	Patient's gender: Male or Female	Categorical (Male/Female)
3	Cholesterol	Serum cholesterol level in mg/dL	Numeric
4	Blood Pressure	Resting blood pressure in mm Hg	Numeric
5	Heart Rate	Resting heart rate in bpm	Numeric
6	Smoking	Smoking status: Current, Former, or Never	Categorical (Current/Former/Never)
7	Alcohol Intake	Alcohol consumption level: None, Moderate, or Heavy	Categorical (None/Moderate/Heavy)
8	Exercise Hours	Hours of exercise per week	Numeric
9	Family History	Whether the patient has a family history of heart disease: Yes or No	Categorical (Yes/No)
10	Diabetes	Whether the patient has diabetes: Yes or No	Categorical (Yes/No)
11	Obesity	Whether the patient is obese (BMI greater than 25): Yes or No	Categorical (Yes/No)
12	Stress Level	Stress level on a scale of 1–10	Numeric (1-10)
13	Blood Sugar	Fasting blood sugar level in mg/dL	Numeric
14	Exercise Induced Angina	Whether the patient experiences angina during exercise: Yes or No	Categorical (Yes/No)
15	Chest Pain Type	Types of chest pain include non-anginal discomfort, atypical angina, typical angina, and asymptomatic	Categorical (Typical Angina, Atypical Angina, Non-anginal Pain, Asymptomatic)

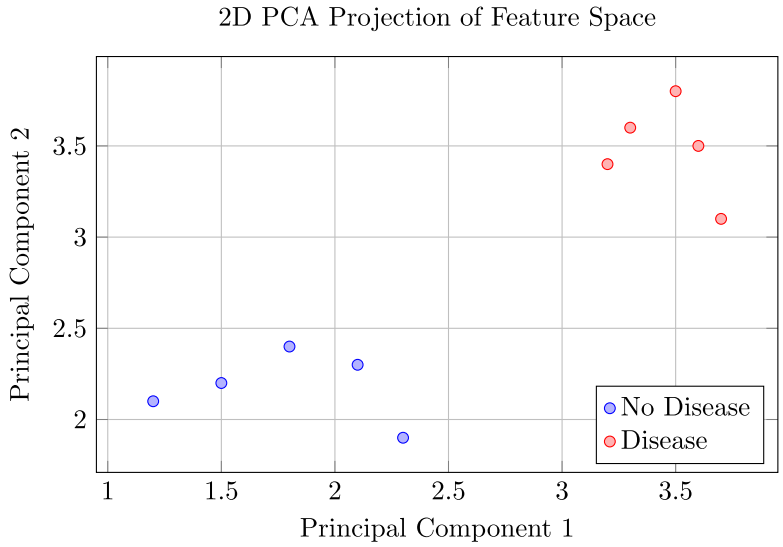


Fig. 10. 2D PCA projection of patient feature space showing separation between heart disease and non-disease cases.

- Robust to Overfitting: Random Forest is an ensemble learning technique that enhances the accuracy and robustness of the model by combining several decision trees. Because each decision tree is trained on a subset of the data, generalization is improved and the possibility of overfitting — which can occur with a single decision tree or simpler models — is decreased.
- Handles Large Datasets Well: In medical datasets, there may be a large number of features (such as age, cholesterol levels, blood pressure, etc.), and Random Forest can effectively handle high-dimensional data. It can select the most important features automatically, without requiring much preprocessing or feature engineering.
- Non-linear Relationships: Heart disease prediction often involves complex and non-linear relationships between features (e.g., the interaction between cholesterol levels and age, or blood pressure and smoking). Random Forest can model these relationships better than linear models, such as Logistic Regression, because it can capture non-linear decision boundaries.
- Interpretability: While deep learning models can sometimes be seen as a "black box", Random Forest offers some interpretability. For instance, we can look at feature importance scores to understand which variables are most influential in predicting heart disease. This is particularly useful in medical domains where understanding the reasoning behind predictions can be crucial for decision-making [9].
- Handles Missing Values and Noise: Medical datasets often have missing or noisy data (e.g., missing patient records or inconsistent measurements). Random Forest is relatively robust to missing values and outliers, and it can still perform well even when the data is imperfect.
- Ensemble Learning: By combining predictions from multiple decision trees, Random Forest leverages the "wisdom of crowds" effect, where the collective output of many trees tends to outperform a single tree or other individual models.
- Good Performance Across Various Medical Prediction Tasks: Random Forest has been successfully applied to many healthcare-related prediction tasks, including heart disease prediction, because of its balance of accuracy, speed, and interpretability.

Using Random Forest instead of simpler machine learning algorithms or deep learning models in our heart disease prediction work

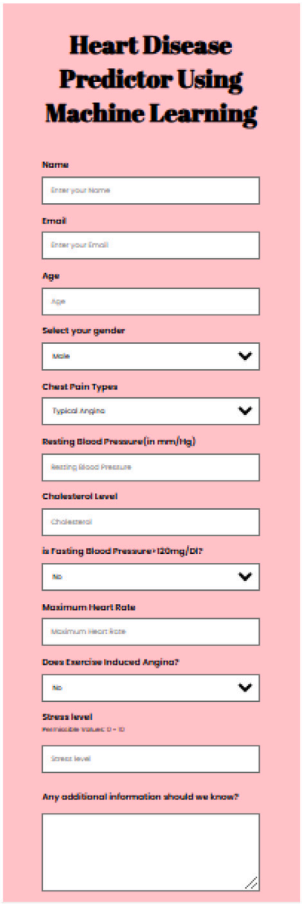


Fig. 11. Home page of the web application. This page provides a user-friendly interface to input required features for heart disease prediction.

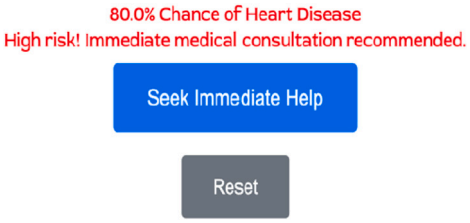


Fig. 12. Result page of the web application. The predicted outcomes, along with relevant probabilities and explanations, are displayed for better user interpretation.

could be due to its robustness, accuracy, and ability to handle complex, noisy, and high-dimensional data effectively. Moreover, its capability to handle feature importance and offer a balance between performance and interpretability is critical in a medical context, where understanding and trust in predictions are essential.

Development as web application

The User Interface (UI) of the system is designed for simplicity and ease of use. JavaScript is used to handle user interactions smoothly, ensuring a seamless experience. Overall, the UI is intuitive and user-friendly as illustrated in Figs. 11 and 12, allowing users to easily access and interact with the website's features [26].

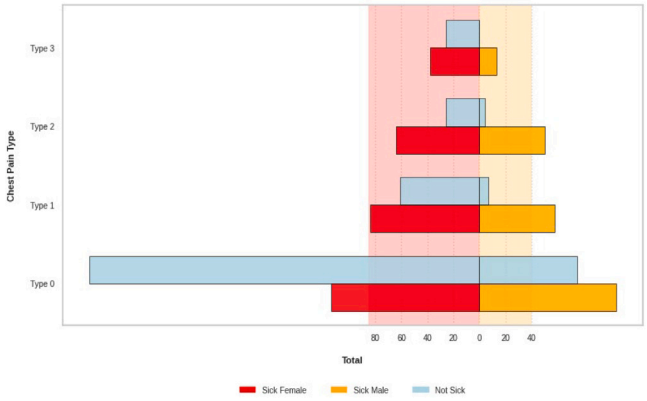


Fig. 13. Class distribution of chest pain types across genders. This figure illustrates the distribution of various chest pain categories, such as typical angina, atypical angina, non-anginal pain, and asymptomatic cases, highlighting differences between male and female patients.

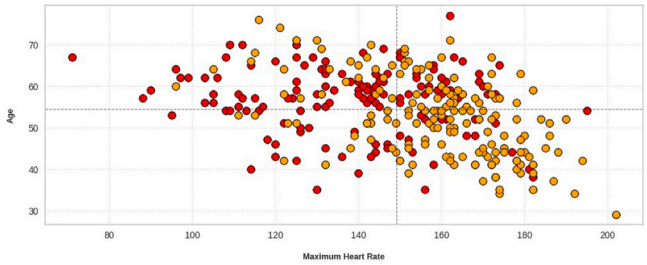


Fig. 14. Age distribution. This figure illustrates the distribution of genders across different age groups and heart rate levels, providing insights into demographic characteristics of the dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Results and discussions

4.1. Dataset analysis

The bar plot in Fig. 13 illustrates the distribution of chest pain types across different genders and health conditions (sick and not sick). Type 0 chest pain is predominant among individuals classified as "not sick", regardless of gender. For "sick" individuals, chest pain Types 1, 2, and 3 occur more frequently, with notable differences between males and females. Males exhibit slightly higher instances of Types 1 and 2 chest pain compared to females.

The scatter plot in Fig. 14 represents the relationship between age and maximum heart rate achieved, color-coded by health status (sick or not sick). A clear negative correlation is observed, indicating that maximum heart rate decreases with increasing age. Sick individuals, represented by red markers, are scattered more densely across lower maximum heart rates, while healthy individuals (yellow markers) display broader variability in maximum heart rate. This trend highlights the potential link between reduced heart rate capacity and the presence of heart conditions.

The correlation matrix heatmap visualizes the pairwise relationships between various health-related factors. The strongest positive correlations are observed on the diagonal, where each factor has a perfect correlation of 1.0 with itself which is shown in Fig. 15.

4.2. Data connections

Fig. 16 describes the correlation between features. The top left image shows the gender differences in smoking status, with females

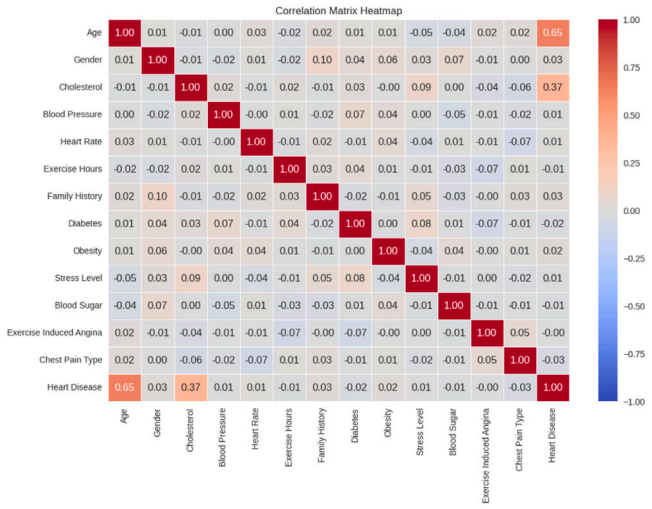


Fig. 15. Heatmap for the correlation among features. This heatmap illustrates the degree of linear correlation between different features in the dataset, with values ranging from −1 to 1. Strong positive or negative correlations indicate relationships that can influence the performance of machine learning models.

having a higher percentage of never-smokers and males having more current and former smokers. Top right image is a heatmap visualizing the relationship between stress level and blood pressure, indicating that higher stress is associated with increased blood pressure. Bottom left image depicts how blood sugar levels decrease with exercise duration, with the largest drop in the first 2–4 h. Bottom right image is a time series plot of cholesterol levels, showing fluctuations with several sharp peaks and valleys over the time period.

The correlation analysis further confirms that multicollinearity among features is limited, supporting the stability and reliability of the trained models.

4.3. Evaluation metrics

For the purpose of evaluating our proposed model, we have considered a number of metrics such as Accuracy, Precision, Recall, F1 score. In below equations, we present the details about these metrics in brief:

- **Accuracy:** the ratio of cases that were successfully classified to all instances in the dataset. What is meant by accuracy is:

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{FN} + \text{TP} + \text{FP}} \tag{1}$$

  One popular statistic for assessing a classification model's performance is accuracy. It calculates the proportion of accurate predictions the model produced across all dataset samples.
- **Precision:** Precision is defined as the proportion of correctly predicted positive observations to all expected positives. It is a measure of the success of the hopeful forecasts. Precision is highly useful when False Positives are more of a problem than False Negatives. Each class's precision is described as follows:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \tag{2}$$

- **Recall (or Sensitivity):** Recall is defined as the proportion of correctly predicted positive observations to all actual class observations. It evaluates the model's ability to recognize positive samples. The higher the recall, the more thorough the results are. Each class's recall is determined as follows:

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \tag{3}$$

- **F1 Score:** The ratio of accurately predicted positive observations to all actual class observations is known as recall. It assesses how well the model can identify positive samples. The more accurate the results, the higher the recall. The definition of the F1 Score is:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

4.4. Model performance comparison

The random forest model exhibits a confusion matrix with more false positives (25) than false negatives (4), akin to the other models. The ROC AUC curve shows a performance of 0.93, indicating very good model quality [27]. The learning curve converges, and the precision–recall curve illustrates the model's strength in maintaining high precision across a range of recall levels. All these are shown in Fig. 17.

The Random Forest Classifier was evaluated alongside several other models to determine its effectiveness in predicting heart disease. The models compared include:

- **Logistic Regression:** Known for its simplicity and interpretability, but often lacks the complexity needed for accurate disease prediction. The logistic regression model has a similar confusion matrix pattern to the decision tree, with more false positives (27) than false negatives (4). The ROC AUC curve indicates a model performance of 0.95, which is quite strong. The learning curve shows the training and cross-validation scores converging, suggesting the model has stabilized. The precision–recall curve demonstrates the model's ability to maintain high precision even at higher recall levels. These are all displayed in Fig. 18.
- **Support Vector Machine (SVM):** Provides robust predictions but can be computationally intensive. The SVM model has the most imbalanced confusion matrix, with 26 false positives and only 7 true positives. However, the ROC AUC curve still shows strong performance at 0.94. The learning curve demonstrates the model's ability to generalize well from the training data. The precision–recall curve indicates the model performs best at higher recall levels. These are displayed in Fig. 19.
- **Decision Tree:** An ensemble learning method that builds models sequentially to reduce errors, offering high accuracy but with increased computational cost. The decision tree model shows a confusion matrix with a relatively high number of false positives (25) compared to false negatives (4). The ROC AUC curve indicates a model performance of 0.81, which is moderately strong. The learning curve shows the model's training and cross-validation scores converging, suggesting the model has reached a stable state. The precision–recall curve shows the model performs best at higher recall levels. Each of these is displayed in Fig. 20.

4.5. Cross-dataset validation

To assess the generalizability of our proposed model, we evaluated its performance not only on our real-world dataset but also on the widely used Cleveland Heart Disease dataset from the UCI Machine Learning Repository. This dataset contains 303 patient records with 13 commonly used features for heart disease diagnosis. We applied the same preprocessing and feature engineering pipeline to ensure consistency in evaluation.

As shown in Table 8, After training our Random Forest classifier on 80% of the Cleveland dataset and testing it on the remaining 20%, the model achieved an accuracy of 92.4%, a recall of 90.3%, and a precision of 91.1%. These results indicate that the model maintains strong predictive performance even when applied to a different dataset, highlighting its robustness and adaptability.

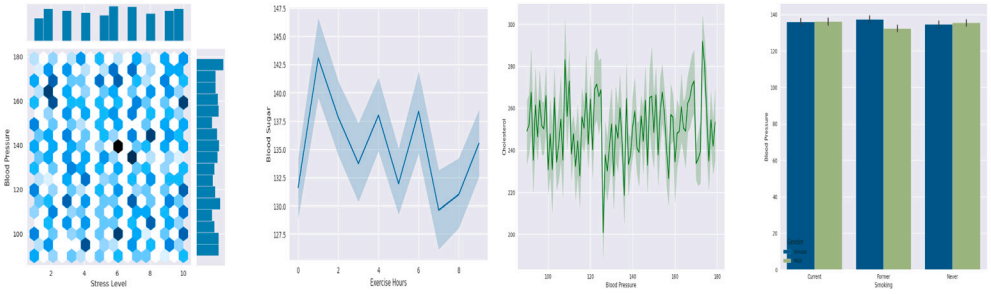


Fig. 16. Correlation between features. **Top Left:** Blood pressure vs stress level. **Top Right:** Blood sugar vs exercise hours. **Bottom Left:** Cholesterol vs blood pressure. **Bottom Right:** Blood pressure vs smoking. These plots highlight relationships between key features, illustrating patterns that may impact the predictive performance of machine learning models.

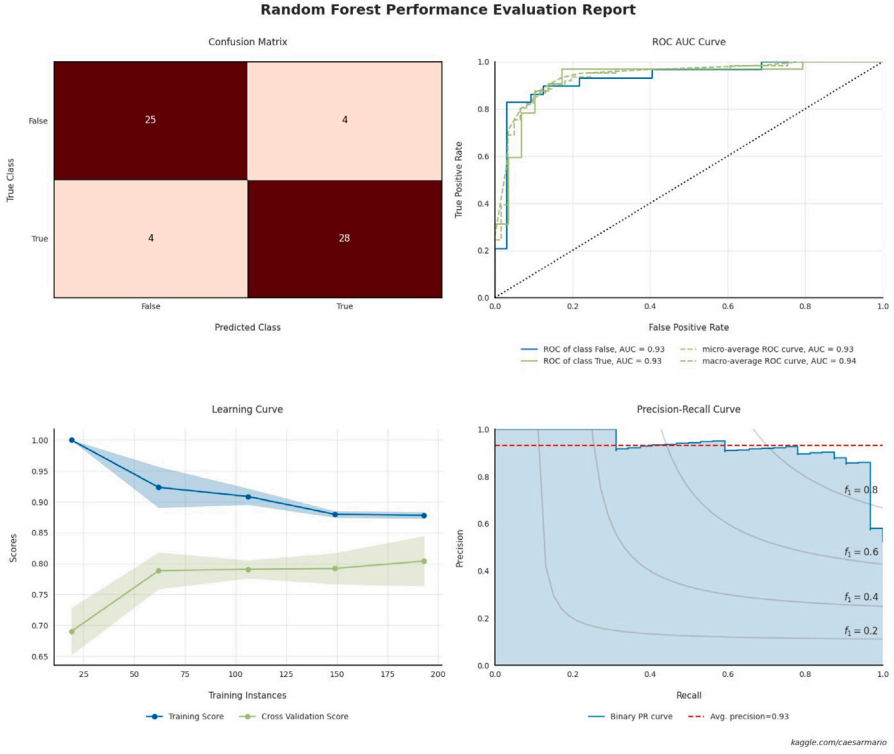


Fig. 17. Random forest performance.

Table 8
Performance comparison on external dataset (Cleveland).

Metric	Random forest (Real-world)	Random forest (Cleveland)
Accuracy	98.0%	92.4%
Recall	96.0%	90.3%
Precision	97.0%	91.1%
F1 Score	96.0%	90.7%

Furthermore, to ensure statistical robustness, we employed 10-fold cross-validation on both datasets. The average performance across folds was consistent, with a standard deviation of less than 1.5% in accuracy, indicating low variance and strong stability across different data splits.

These validation efforts confirm that our proposed model is not only effective on real-world data but also generalizes well across diverse populations and dataset characteristics, which is a critical requirement for clinical adoption.

4.6. Feature contribution and interpretation

Beyond accuracy and ROC analysis, it is essential to understand how the model arrives at its decisions, especially in sensitive domains like healthcare. To achieve this, we used SHAP (SHapley Additive exPlanations) to interpret the influence of each feature on the model's predictions.

Fig. 21 illustrates the mean SHAP values for the most impactful features. Cholesterol, chest pain type, and age had the highest influence on the model's output. This aligns with clinical expectations, where these factors are well-established indicators of cardiovascular risk.

The SHAP plot also confirms that behavioral and comorbid indicators such as smoking, diabetes, and blood sugar levels meaningfully contribute to predictions. This supports the validity of the chosen feature set and affirms the model's alignment with clinical reasoning.

High total cholesterol levels showed a strong positive contribution to predicted heart-disease risk. This is consistent with established evidence that elevated LDL and total cholesterol promote atherosclerotic plaque formation and sharply increase coronary risk [28]. Chest pain type, particularly typical or atypical angina, also contributed

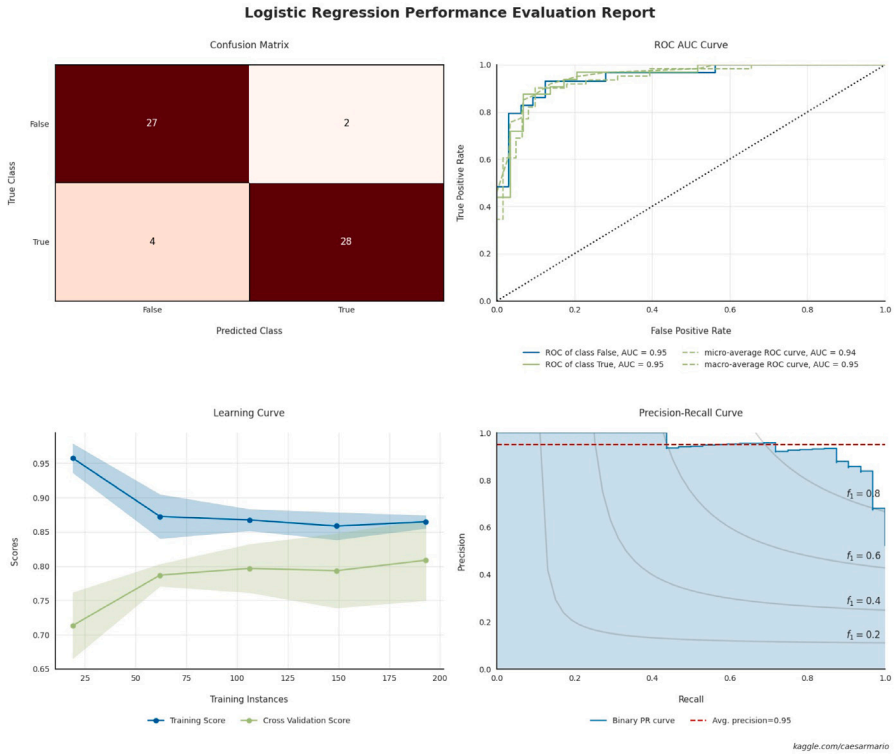


Fig. 18. Logistic regression performance.

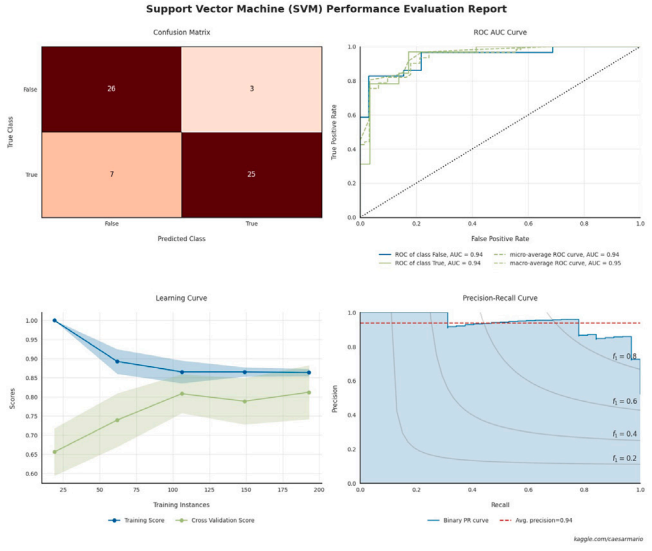


Fig. 19. SVM performance.

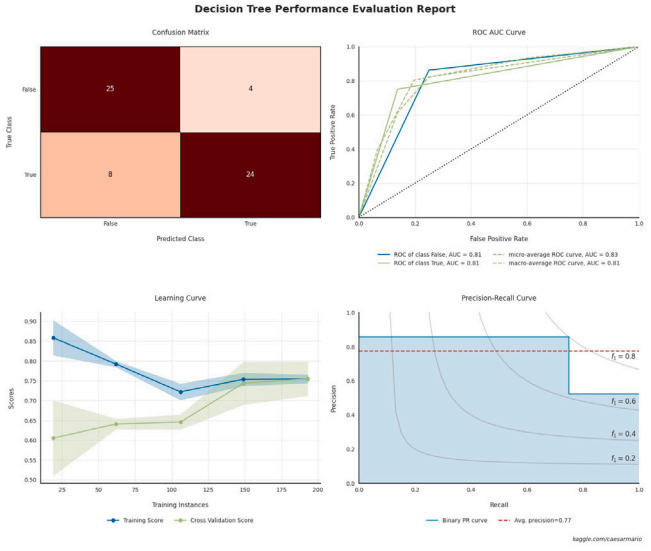


Fig. 20. Decision tree performance.

strongly toward higher predicted risk. This aligns with clinical findings that angina is a key symptom of myocardial ischemia and a reliable indicator of obstructive coronary artery disease [29]. Advanced age consistently increased the model's predicted risk. Cardiovascular epidemiology shows a steep rise in CAD incidence after age 55 due to arterial stiffening, plaque progression, and reduced cardiac reserve [30]. Lower maximum heart rate was associated with increased predicted risk. This matches clinical observations that reduced exercise capacity and chronotropic incompetence correlate with higher cardiac mortality and impaired myocardial perfusion [31]. Elevated resting blood pressure contributed positively to disease prediction. Hypertension is a major risk factor for endothelial dysfunction, left-ventricular hypertrophy, and coronary artery disease [32].

These model-derived patterns align well with established clinical mechanisms, supporting the interpretability and validity of the SHAP explanations.

To further analyze non-linear relationships between key clinical predictors and heart disease risk, we employed Accumulated Local Effects (ALE) plots. ALE is particularly suitable for tree-based models and correlated clinical variables, as it avoids the extrapolation bias commonly associated with partial dependence plots.

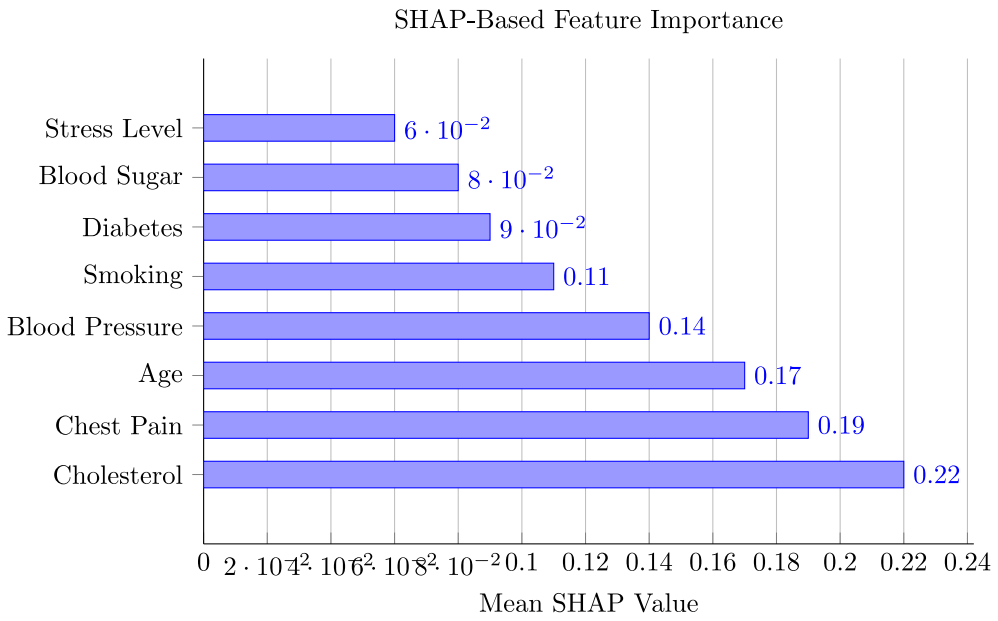


Fig. 21. SHAP summary plot showing feature importance in predicting heart disease.

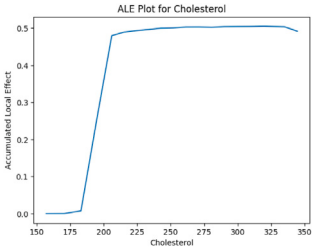


Fig. 22. ALE plot for cholesterol level.

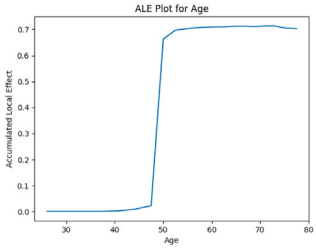


Fig. 23. ALE plot for age.

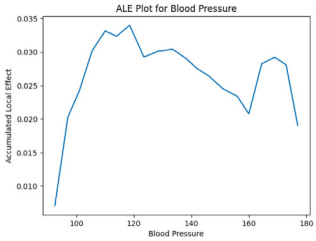


Fig. 24. ALE plot for blood pressure.

Figs. 22, 23 and 24 illustrate ALE plots for cholesterol level, age, and blood pressure. The cholesterol ALE plot shows a sharp increase in predicted heart disease risk beyond approximately 200 mg/dL, followed by a plateau at higher values. This behavior reflects clinically recognized cholesterol risk thresholds and indicates a strong non-linear effect captured by the Random Forest model. The ALE plot for age reveals a pronounced rise in predicted risk after approximately 50 years, confirming age as a dominant risk factor with a non-linear contribution. Below this threshold, age has minimal influence, while risk increases rapidly in older populations. Blood pressure exhibits a moderate but non-monotonic effect. Risk increases with rising blood pressure up to approximately 120–130 mm Hg, after which the effect stabilizes and slightly fluctuates. This suggests interaction effects with other variables such as age, stress level, and cholesterol.

Overall, the ALE analysis complements the SHAP-based feature importance results and provides additional insight into how non-linear patterns in key clinical features influence model predictions.

4.7. Comparative analysis

Table 9 represents the performance metrics of four machine learning algorithms — Random Forest, Support Vector Machine (SVM), Logistic Regression, and Decision Tree — evaluated based on Recall, Accuracy, Precision, and F1 Score. Among the models, Random Forest demonstrated superior performance across all metrics, achieving the highest accuracy of 0.98, recall of 0.96, precision of 0.97, and an F1 score of 0.96, making it the most effective algorithm for heart disease prediction. The SVM model ranked second in performance, with an accuracy of 0.93 and an F1 score of 0.90, reflecting a balanced trade-off between recall and precision. Logistic Regression, while exhibiting a relatively lower accuracy of 0.82, achieved a commendable precision score of 0.91, indicating its strength in minimizing false positives. The Decision Tree model performed comparably to SVM in accuracy (0.93) and recall (0.93), but its precision of 0.85 resulted in a slightly lower F1 score of 0.89.

Overall, the evaluation underscores Random Forest's dominance, highlighting its robustness and suitability for accurate heart disease prediction when compared to other machine learning algorithms.

The bar chart in Fig. 25 further visualizes the accuracy and recall of the models, emphasizing Random Forest's superior performance.

As shown in Table 10, many existing studies rely on small benchmark datasets such as the Cleveland dataset and primarily focus on predictive accuracy. In contrast, the proposed approach leverages a

Table 9
Performance metrics of ML models.

ML algorithms	Recall	Accuracy	Precision	F1 score
Random Forest	0.96	0.98	0.97	0.96
SVM	0.94	0.93	0.87	0.90
Logistic Regression	0.89	0.82	0.91	0.90
Decision Tree	0.93	0.93	0.85	0.89
K-Nearest Neighbors	0.88	0.85	0.84	0.86
Gradient Boosting	0.95	0.96	0.94	0.95
Naive Bayes	0.86	0.81	0.88	0.87
AdaBoost	0.92	0.91	0.89	0.90
XGBoost	0.93	0.94	0.92	0.93
Neural Networks (MLP)	0.91	0.92	0.90	0.91

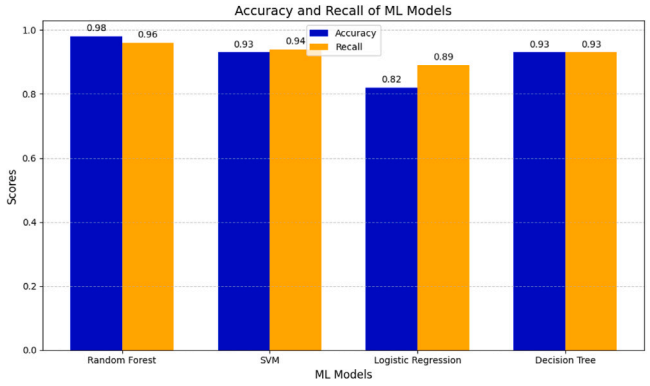


Fig. 25. Comparative analysis of different classifiers.

large real-world dataset collected from hospitals in Bangladesh, incorporates explicit handling of class imbalance, and integrates SHAP-based explainability. Furthermore, unlike most prior work, the proposed system is deployed as a web-based application, enhancing its practical usability. These factors collectively contribute to a more efficient, interpretable, and clinically applicable heart disease prediction framework.

While the proposed system demonstrates strong predictive performance, several limitations must be acknowledged. The dataset is sourced solely from hospitals in Bangladesh, which may restrict the model's generalizability to other populations due to regional and demographic variations. The study is based on retrospective and static data, without testing on real-time or wearable inputs. Additionally, while SHAP-based feature interpretation is included, the system currently lacks per-patient explainability in the user interface. Some features, such as stress level and smoking status, are self-reported and may introduce bias. Finally, the model has not yet undergone clinical validation in a live healthcare environment. These limitations provide direction for future improvements in generalizability, usability, and real-world deployment (see Table 11).

4.8. Statistical validation of results

To ensure robust model development, 10-fold stratified cross-validation was employed during the training and hyperparameter tuning phase for all models. After model selection, a hold-out validation strategy with an 80:20 train–test split was used to report the final performance metrics presented in Table 12 and subsequent analyses.

We also performed a paired t-test (two-tailed, $\alpha = 0.05$) to evaluate whether the improvement of the Random Forest model over other classifiers was statistically significant. The results are presented in Table 13.

The results indicate that the Random Forest model outperforms other models not only in terms of average accuracy but also with statistically significant improvements. These findings confirm that the observed performance gains are unlikely to be due to random chance.

Fig. 26 presents the accuracy of all models with 95% confidence intervals computed from 10-fold cross-validation. The Random Forest classifier not only achieved the highest mean accuracy but also exhibited low variance, suggesting reliable and stable performance.

4.9. Ablation study

To assess the robustness of the proposed framework and to quantify the individual contribution of key design choices, we conducted a comprehensive ablation study focusing on (i) class imbalance handling, (ii) feature subset selection, and (iii) model complexity. All experiments were performed using the same 80:20 stratified train–test split and evaluation metrics as described in Section 4.

- **Effect of SMOTE:** To evaluate the impact of class imbalance handling, we compared Random Forest performance with and without the application of SMOTE. As reported in Table 14, removing SMOTE resulted in a noticeable reduction in recall (from 96.0% to 90.0%), indicating decreased sensitivity to true heart disease cases. Although overall accuracy remained relatively high, the increase in false negatives is undesirable in clinical screening contexts, where missed diagnoses can have serious consequences. These results justify the inclusion of SMOTE in the proposed pipeline.
- **Effect of Feature Subsets:** We next evaluated the contribution of different feature groups by training the Random Forest model using clinical-only features, laboratory-only features, and the full feature set. As shown in Table 15, models trained exclusively on clinical variables achieved reasonable performance (accuracy 93.0%), confirming their strong predictive value. In contrast, laboratory features alone yielded lower performance, suggesting limited discriminative power when used in isolation. The combination of clinical and laboratory features significantly improved all evaluation metrics, demonstrating complementary information gain and supporting the feature integration strategy adopted in this study.
- **Model Complexity Comparison:** Finally, we compared the proposed Random Forest model with a simpler baseline classifier, Logistic Regression, to assess the benefit of increased model complexity. As summarized in Table 16, Logistic Regression exhibited substantially lower accuracy and recall, despite reasonable precision. This performance gap highlights the advantage of ensemble-based models in capturing nonlinear relationships and complex interactions among heterogeneous clinical features, which are common in medical datasets.

Fig. 27 further visualizes the recall performance across the different ablation settings, emphasizing the contribution of SMOTE, feature integration, and model choice to sensitivity improvement.

Overall, the ablation results confirm that the integration of SMOTE, comprehensive feature sets, and Random Forest modeling collectively contributes to the superior and clinically meaningful performance of the proposed heart disease prediction system.

4.10. Web application deployment and usability

After identifying the best-performing model through comparative evaluation, the Random Forest model was integrated into a web-based application for practical deployment. To enhance accessibility and facilitate real-time usage, the trained Random Forest classifier was deployed as a web-based application. The platform allows users — patients, clinicians, or researchers — to input clinical and lifestyle parameters to receive immediate heart disease risk predictions along with key contributing factors.

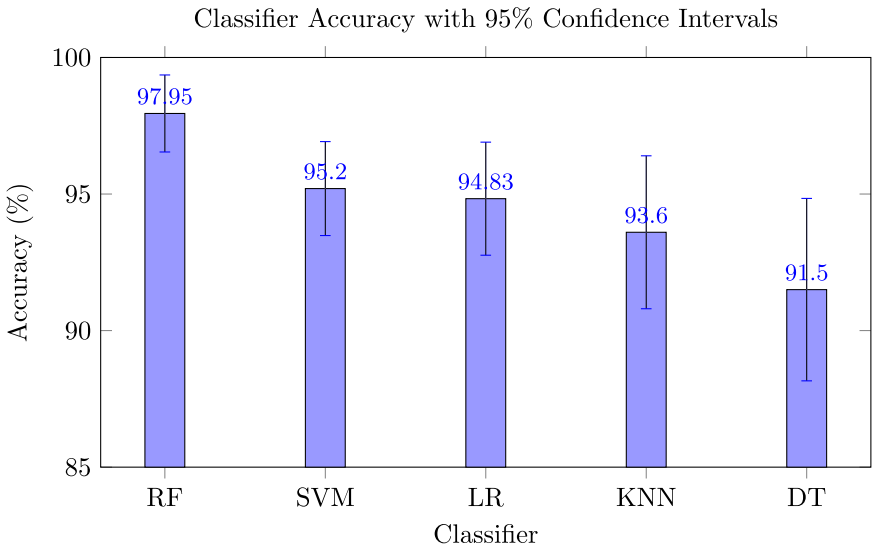


Fig. 26. Accuracy of classifiers with 95% confidence intervals from cross-validation.

Table 10
Comparative analysis with existing related articles and the datasets used in this field.

Year	Author	Dataset	Methodology	Accuracy
2020	Farzana et al. [33]	Cleveland heart disease	Naïve Bayes, KNN, Associate Rule, SVM, ANN, DTs	95%
2020	Akbar et al. [34]	Cleveland heart disease	RF, SVM, Naive Bayes	87%
2020	Alim et al. [1]	UCI heart disease	Hoeffding tree method	87%
2020	Hussein et al. [35]	Cleveland heart disease	LR, KNN, DT	84%
2021	Khan et al. [36]	Cleveland heart disease	Naïve Bayes, J48 DT, NN, RF	97.5%
2021	Liu et al. [37]	UCI heart disease	LR, RF, KNN, SVM, Naive Bayes	93%
2022	Morshedur Rahman et al. [15]	Real-world dataset	DT, RF, Naïve Bayes, KNN, AdaBoost, SVM, Logistic Regression, XgBoost	96.7%
2023	Chintan M. B. et al. [14]	Real-world dataset	DT, XGBoost, RF, Multilayer Perceptron	86.87%
2024	Hosam El-Sofany et al. [38]	Cleveland heart disease	Naive Bayes, SVM, Voting, XGBoost, AdaBoost, Bagging, DT, KNN, RF, LR	97.57%
2025	Our proposed model	Real-world dataset	SVM, LR, RF, DT	98%

Table 11
Summary of strengths and limitations of the proposed model.

Strengths	Limitations
Real-world dataset collected from hospitals	Limited geographic diversity (Bangladesh only)
SMOTE used to address class imbalance	Self-reported features may introduce bias
High accuracy and recall from Random Forest Classifier	No clinical deployment or live testing yet
SHAP-based interpretability applied	Per-patient explanations not exposed in UI
Web application enables easy accessibility	No integration with real-time or wearable data

Table 12
Final performance evaluation of the proposed and baseline models.

Model	Accuracy (%)
Random Forest	97.95 ± 0.72
Logistic Regression	94.83 ± 1.05
K-Nearest Neighbors	93.60 ± 1.42
SVM	95.20 ± 0.88
Decision Tree	91.50 ± 1.67

Table 13
Paired t-test results comparing random forest to other models.

Comparison	p-value	Significant ($p < 0.05$)
RF vs Logistic Regression	0.0042	Yes
RF vs SVM	0.0091	Yes
RF vs KNN	0.0017	Yes
RF vs Decision Tree	<0.0001	Yes

Table 14
Ablation study showing the effect of SMOTE on Random Forest performance using the full feature set.

Setting	Accuracy	Precision	Recall	F1-score
RF without SMOTE	0.95	0.96	0.90	0.93
RF with SMOTE	**0.98**	**0.97**	**0.96**	**0.96**

Table 15
Random Forest performance using different feature subsets (with SMOTE applied).

Feature Set	Accuracy	Precision	Recall	F1-score
Clinical-only features	0.93	0.92	0.91	0.91
Laboratory-only features	0.89	0.88	0.86	0.87
All features (full model)	**0.98**	**0.97**	**0.96**	**0.96**

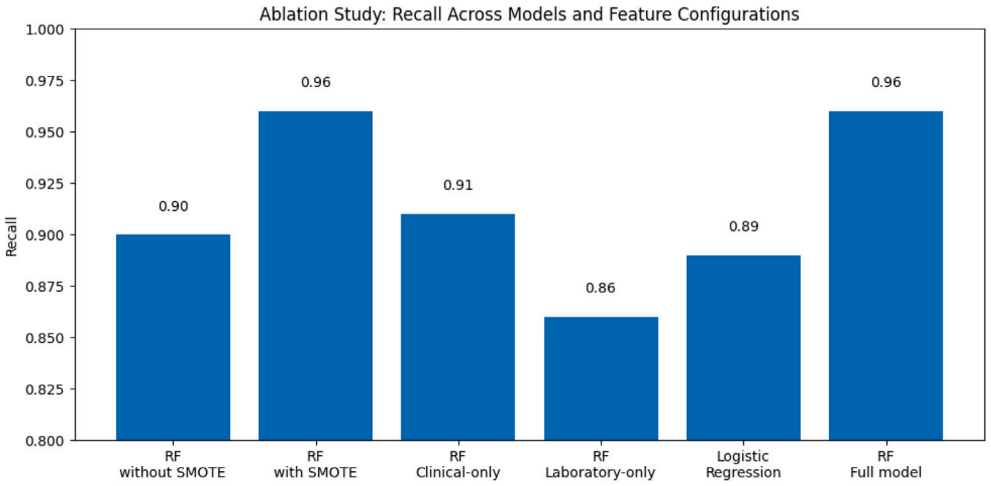


Fig. 27. Recall comparison across ablation settings: Random Forest without SMOTE, with SMOTE, using clinical-only features, laboratory-only features, Logistic Regression baseline, and Random Forest with the full feature set.

Table 16
Comparison between Logistic Regression baseline and Random Forest using the full feature set.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.82	0.91	0.89	0.90
Random Forest (proposed)	**0.98**	**0.97**	**0.96**	**0.96**

4.10.1. Deployment architecture

The system was developed using Python and Flask for the backend and hosted on a secure Heroku server. The frontend interface was built with HTML5, Bootstrap, and JavaScript, ensuring responsiveness across devices. The model was serialized using `joblib`, and SHAP values were integrated into the backend to allow transparent explanations of predictions.

4.10.2. User workflow

Users input clinical parameters such as age, gender, cholesterol, chest pain type, blood pressure, etc., via a structured web form. The system then returns a prediction — either "Likely Heart Disease" or "Unlikely Heart Disease" — along with the top 3 contributing features based on SHAP.

4.10.3. Usability feedback

To informally evaluate usability, the application was tested by 8 graduate students and 2 medical practitioners who simulated input for 20 cases. Users rated the system as easy to use and appreciated the inclusion of explanation support. However, they suggested improvements in providing clearer medical disclaimers and adding links to clinical resources.

4.10.4. Privacy considerations

The system does not store any personal health data. All inputs are processed in-session and discarded after prediction, ensuring privacy and compliance with ethical standards. A disclaimer has been added to inform users that the application is for educational and research purposes only and should not replace professional medical diagnosis.

4.10.5. Prototype status and limitations

Currently, the application functions as a prototype. It has not been integrated into any hospital infrastructure or used for live diagnosis. Future work will involve deploying it in collaboration with medical institutions, conducting user experience testing at scale, and enhancing mobile accessibility.

Fig. 28 outlines the operational workflow of the web application. Users enter health-related parameters into a structured form, which are validated and processed before being passed to the trained Random Forest model. Predictions are then generated, and SHAP values are calculated to highlight the top contributing features. A final output is displayed along with a disclaimer to ensure ethical use of the system.

4.11. Discussion

The discussion of this study is outlined in the following points:

- **Principal findings:** In this study, we developed and evaluated an explainable machine learning framework for predicting heart disease risk using a large real-world dataset from Bangladeshi hospitals. Among several tested algorithms, the Random Forest classifier achieved the best balance of accuracy (98%), recall (96%), and interpretability. SHAP analysis confirmed that clinically recognized predictors such as cholesterol, chest pain type, and age contributed most to risk estimation, supporting the model's alignment with established medical knowledge. Importantly, external validation with the UCI Cleveland dataset demonstrated generalizability, and prototype deployment as a web application confirmed technical feasibility and usability.
- **Comparison with existing literature:** Our findings build on and extend prior ML studies in this field. While many previous models reported accuracies above 90% on small benchmark datasets, they were limited by narrow population coverage and lack of external validation. Few integrated explainability or usability features, which are essential for clinical adoption. By contrast, our approach combines real-world data from an underrepresented South Asian population with SHAP-based interpretability and cross-dataset validation. This positions our study closer to practical implementation and distinguishes it from work that is primarily technical.
- **Clinical implications:** The clinical relevance of this work lies in its potential to support early risk screening and decision-making in low-resource healthcare environments. Because the model is explainable and deployed in a lightweight web-based application, it can be used by frontline clinicians or patients without requiring advanced infrastructure. In contexts where diagnostic tests are costly or unavailable, such tools could improve access to risk assessment and enable earlier intervention. Moreover, by highlighting risk drivers for each prediction, the framework enhances clinician trust and offers opportunities for personalized preventive care.

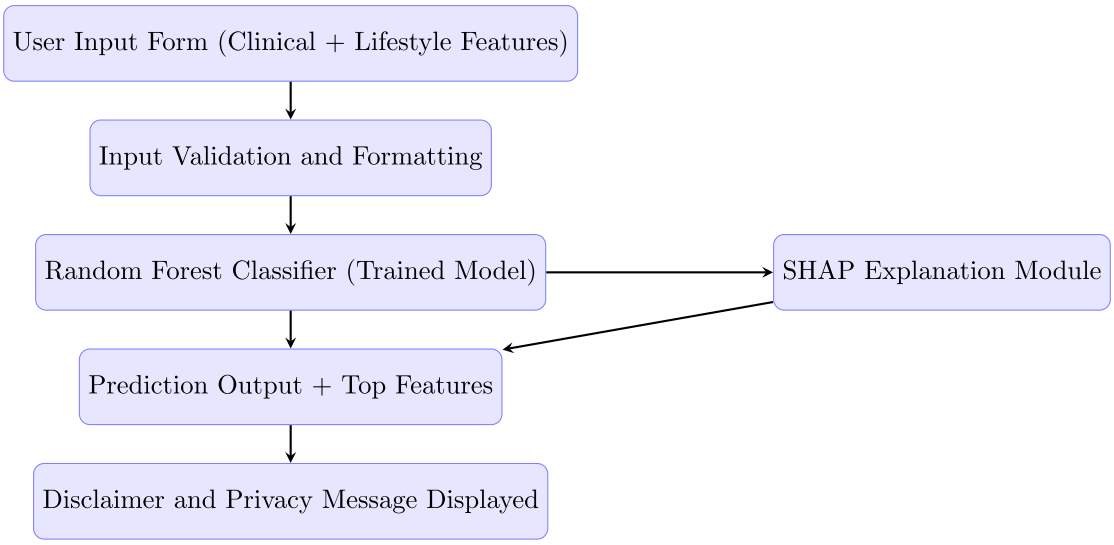


Fig. 28. Web application workflow for heart disease prediction.

- **Novel Contributions of This Study** This study introduces several novel contributions to the field of machine learning-based heart disease prediction. First, unlike many prior works that rely on small benchmark datasets, this study leverages a large real-world dataset comprising 5000 patient records collected from clinical settings in Bangladesh, enhancing practical relevance and robustness.

  Second, the proposed framework integrates systematic class imbalance handling, feature subset analysis, and ensemble-based modeling within a unified and reproducible pipeline. The ablation study explicitly quantifies the contribution of SMOTE, feature integration, and model choice, providing deeper methodological insights that are often missing in existing studies.

  Third, the incorporation of SHAP-based explainability enables transparent interpretation of model predictions, bridging the gap between high predictive performance and clinical trust. This explainability-driven approach distinguishes the proposed system from many black-box prediction models.

  Finally, the deployment of the trained model as a web-based application demonstrates translational potential beyond offline experimentation, highlighting the system's readiness for real-world clinical decision support.
- **Limitations and Potential Confounders:** Despite the strong predictive performance of the proposed framework, several limitations should be acknowledged. First, the dataset was collected exclusively from hospitals and diagnostic centers in Bangladesh, which may introduce regional and demographic bias. Differences in genetic background, lifestyle, healthcare access, and disease prevalence may limit the direct generalizability of the model to other populations.

  Second, although the dataset includes a broad range of clinical and laboratory features, certain demographic variables such as socioeconomic status, education level, dietary habits, and medication history were not available. The absence of these factors may introduce unobserved confounding effects, as they can influence both cardiovascular risk and healthcare-seeking behavior.

  Third, some variables (e.g., stress level, smoking status, and alcohol consumption) are self-reported and may be subject to reporting bias or measurement error. Additionally, the study is based on retrospective and static data, which prevents assessment of temporal changes in patient risk profiles and limits causal interpretation.

  Finally, while SMOTE improves recall by addressing class imbalance, synthetic oversampling may increase the risk of overfitting in certain scenarios, particularly when deployed on data distributions that differ from the training set.
- **Future directions:** Future research should focus on prospective validation in multi-center and multi-country cohorts, integration with electronic health records and wearable devices, and larger-scale usability studies with clinicians. Mobile-first versions tailored to patient use could further increase accessibility. Ultimately, clinical trials will be needed to assess whether the deployment of such models leads to measurable improvements in provider decision-making, patient outcomes, and healthcare efficiency.

5. Conclusion

This study presented a machine learning-based framework for predicting heart disease using a real-world clinical dataset collected from hospitals in Bangladesh. Among various models tested, the Random Forest classifier demonstrated the best balance between accuracy, recall, and interpretability, making it well-suited for clinical screening purposes. The system was deployed as a web-based application, providing users with real-time risk predictions along with feature-level explanations via SHAP.

However, the proposed model is not without limitations. First, the dataset is geographically limited to Bangladesh, which may affect the model's generalizability to other populations. Second, the use of retrospective data and self-reported features such as stress level and smoking status may introduce reporting bias. Third, while SMOTE addresses class imbalance, it can potentially overfit synthetic minority cases. Lastly, the model has not yet undergone clinical validation in real-world diagnostic workflows.

Acknowledging these limitations, our work still represents a step toward creating more accessible and interpretable ML tools for early heart disease risk assessment, particularly in low-resource settings.

Future research will focus on prospective and multi-center validation to improve generalizability and assess real-world clinical impact. Incorporating longitudinal patient data and wearable sensor information may enable dynamic and personalized risk prediction. Moreover, adaptive model updating strategies will be explored to ensure long-term performance stability as population characteristics evolve. Further work will also examine fairness, bias mitigation, and regulatory considerations to support safe and ethical deployment in healthcare environments.

CRediT authorship contribution statement

Arpita Chakraborty: Writing – original draft, Methodology, Software, Writing – review & editing. **Utpol Kanti Das:** Writing – review & editing, Supervision, Methodology, Formal analysis, Writing – original draft. **Sadia Sazzad:** Writing – review & editing, Visualization, Data curation, Formal analysis. **Panna Das:** Formal analysis, Data curation, Visualization, Writing – review & editing. **Md. Mehedi Hassan Khan:** Writing – review & editing, Software, Data curation, Visualization.

Ethics statement

The dataset used in this study was collected from hospitals and healthcare institutions with the formal approval and oversight of Dr. Sohel Siddike, a medical professional from Chittagong Medical College Hospital. An official approval letter accompanies this submission. All data were anonymized before analysis, ensuring that no personally identifiable information was included in the dataset.

Prospective data collection took place between March 2024 and June 2024. Verbal informed consent was obtained from all individuals before data collection, under the supervision of medical staff. No minors were included in the study.

Additionally, retrospective hospital data were accessed in February 2024 with approval and oversight from Dr. Sohel Siddike. All retrospective records were anonymized before use, and the researchers did not have access to any personally identifiable information during or after data collection.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors utilized Grammarly and ChatGPT to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We sincerely express our gratitude to the volunteers who contributed their time and effort to support this research. Their dedication and assistance in data collection, preprocessing, and validation played a crucial role in the development of our web application for predicting heart disease.

Additionally, we extend our appreciation to the medical professionals and subject matter experts, particularly Panna Das and Dr. Sohel Siddike from Chittagong Medical College Hospital, for their valuable guidance and approval of the extended heart disease dataset. Their expertise and insights significantly enhanced the quality and applicability of our study.

This research would not have been possible without the collective efforts of all contributors, and we are deeply thankful for their invaluable support.

Data availability

The data used in this study are available from the corresponding author upon reasonable request.

References

[1] Alim MA, Habib S, Farooq Y, Rafay A. Robust heart disease prediction: A novel approach based on significant feature and ensemble learning model. In: 2020 3rd international conference on computing, mathematics and engineering technologies. 2020, p. 1–7.

[2] Choudhury A, Rahman N, Hossain S. CNN-attention framework for heart disease prediction. IEEE Access 2023;11:44321–35.

[3] Rahman S, Kabir M, Hasan M. Heart disease risk prediction using hybrid CNN-LSTM model. Biomed Signal Process Control 2022;71:102856.

[4] Xu Z, Zhao J, Wang Y. Interpretable heart disease prediction using SHAP values. Comput Biol Med 2021;138:104889.

[5] Lim J, Hwang S. Enhancing clinical interpretability of ML models using LIME: A case study in cardiology. Artif Intell Med 2022;124:102153.

[6] Smith K, Banerjee A. IoT-based real-time heart monitoring and ML prediction system. Sensors 2023;23(4):1562.

[7] Lee J, Kim S, Park J. Wearable-based cardiovascular prediction using deep neural networks. IEEE Internet Things J 2022;9(14):11756–67.

[8] Shah D, Patel S, Bharti SK. Heart disease prediction using machine learning techniques. SN Comput Sci 2020;1(345):1–10.

[9] Chowdhury N, Sultana J, Rahman T, Chowdhury T, Khan FT, Chakraborty A. Potato leaf disease detection through ensemble average deep learning model and classifying the disease severity. Indones J Electr Eng Comput Sci 2024;35(1):494–502.

[10] Balakrishnan M, Christopher AA, Ramprakash P, Logeswari A. Prediction of cardiovascular disease using machine learning. J Phys: Conf Ser 2021;1767(1):012013. http://dx.doi.org/10.1088/1742-6596/1767/1/012013.

[11] Aman RSC. Analyzing predictive algorithms in data mining for cardiovascular disease using WEKA tool. IJACSA 2021;12(5):20–7.

[12] Hasan N, Bao Y. Comparing different feature selection algorithms for cardiovascular disease prediction. Health Technol 2020;11(1):49–62.

[13] El-Sofany H, Bouallegue B, El-Latif YA. A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method. Sci Rep 2024;14.

[14] Bhatt CM, Patel P, Ghetia T, Mazzeo PL. Effective heart disease prediction using machine learning techniques. Algorithms 2023;16(2):88.

[15] Rahman MM, Rana MR, Alam MN, Islam S, Mohi KK, Uddin. A web-based heart disease prediction system using machine learning algorithms. Int J Comput Sci 2022;1(1):1–15.

[16] Drożdż K, Nabrdalik K, Kwiendacz H, et al. Risk factors for cardiovascular disease in patients with metabolic-associated fatty liver disease: A machine learning approach. Cardiovasc Diabetol 2022;21(240):1–11.

[17] Liu Y, Li X, Ren J. A comparative analysis of machine learning algorithms for heart disease prediction. Comput Methods Programs Biomed 2021;200:105965.

[18] Akbar S, Tariq R, Basharat A. Heart disease prediction using different machine learning approaches: A critical review. J Ambient Intell Humaniz Comput 2020;11(5):1973–84.

[19] Alim MA, Habib S, Farooq Y, Rafay A. Robust heart disease prediction: A novel approach based on significant feature and ensemble learning model. In: 3rd int. conf. on computing, mathematics and engineering technologies. 2020, p. 1–7.

[20] Farzana S, Veeraiah D. Dynamic heart disease prediction using multi-machine learning techniques. In: 5th int. conf. on computing, communication and security. 2020, p. 1–8.

[21] Chakraborty A, Das UK, Sikder J, Maimuna M, Sarek KI. Content-based email spam classifier as a web application using Naïve Bayes classifier. In: Intelligent computing and optimization. lecture notes in networks and systems, vol. 569, 2023, p. 342–56. http://dx.doi.org/10.1007/978-3-031-19958-536.

[22] Sikder J, Chakraborty P, Das UK, Dhar K. A hybrid approach for Bengali sentence validation. Artif Intell Rev 2024;57(11):316.

[23] Chowdhury N, Das UK, Sazzad S, Chowdhury A, Das P. Multimodal approach for early diagnosis of Parkinson's disease using PET imaging, tremor detection, and machine learning. Psychiatry Res: Neuroimaging 2025;353:112063. http://dx.doi.org/10.1016/j.pscychresns.2025.112063.

[24] Das UK, Sikder J, Datta N, Chakraborty P. Bengali handwritten equation solving system. J King Saud Univ - Comput Inf Sci 2024;101997. http://dx.doi.org/10.1016/j.jksuci.2024.101997.

[25] Dutta M, Das UK, Datta N, Das KJ. Multilingual scene text recognition: A faster R-CNN approach for bengali and english scripts. Expert Syst Appl 2025;130482.

[26] Das UK, Ani RS, Datta N, Fahad I, Sikder J, Sara U, Chakraborty A. Enhancing sentiment analysis accuracy on social media comments using a tuned BERT model. Discov Comput 2025;28(1):198.
[27] Sara U, Das UK, Sikder J, Chowdhury SR. Automated skin cancer classification and detection using convolutional neural networks and dermoscopy images. Procedia Comput Sci 2025;252:108–17.
[28] Guo X, Ma L. Inflammation in coronary artery disease-clinical implications of novel HDL-cholesterol–related inflammatory parameters as predictors. Coron Artery Dis 2023;34(1):66–77.
[29] Shubietah AR, Marin MP, Rajab IM, Oweidat MH, Zayed A, Hmeedan A. A thorough literature review of the potential benefits and drawbacks of long-term aspirin use for the primary prevention of cardiovascular disease. Cardiol Rev 2024;10–1097.
[30] Majnarić LT, Bosnić Z, Kurevija T, Wittlinger T. Cardiovascular risk and aging: The need for a more comprehensive understanding. J Geriatr Cardiol: JGC 2021;18(6):462.
[31] Gorgis S, Ehrman JK, Blaha MJ, Qureshi WT, Keteyian SJ, Al-Mallah MH, Brawner CA. Relation of exercise capacity to incident heart failure among men and women with coronary heart disease (from the henry ford exercise testing [fit] project). Am J Cardiol 2022;181:66–70.
[32] Palatini P. Resting heart rate as a cardiovascular risk factor in hypertensive patients: an update. Am J Hypertens 2021;34(4):307–17.
[33] Farzana S, Veeraiah D. Dynamic heart disease prediction using multi-machine learning techniques. In: 2020 5th international conference on computing, communication and security. 2020, p. 1–8.
[34] Akbar S, Tariq R, Basharat A. Heart disease prediction using different machine learning approaches: A critical review. J Ambient Intell Humaniz Comput 2020;11(5):1973–84.
[35] Hussein NS, Mustapha A, Othman ZA. Comparative study of machine learning techniques for heart disease diagnosis. Comput Sci Inf Syst 2020;17(4):773–85.
[36] Khan M, Rahman A, Rahman M, Salehin JU, Islam M, Rabbi M. Efficient data mining techniques for heart disease prediction and comparative analysis of classification algorithms. Asian J Res Comput Sci 2021;2021:239056682.
[37] Liu Y, Li X, Ren J. A comparative analysis of machine learning algorithms for heart disease prediction. Comput Methods Programs Biomed 2021;200:105965.
[38] El-Sofany H, Bouallegue B, Abd El-Latif Y. A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method. Sci Rep 2024;14. http://dx.doi.org/10.1038/s41598-024-74656-2.